

Prediction-Powered Data Fusion for Treatment Effect Estimation

Yonghan Jung

Working draft, September 9, 2026

Randomized controlled trials identify causal effects under transparent design assumptions, but their sample sizes can make average and conditional treatment-effect estimation imprecise. Observational studies are often larger and can yield accurate outcome predictions, but those predictions need not be causal. We develop a model-agnostic prediction-powered framework that keeps the randomized trial as the causal anchor while using observational outcome models and covariates through two tunable channels. The coefficient λ changes the outcome regression inside an RCT-valid pseudo-outcome, and ω introduces a prediction-powered control variate centered across the two samples.

For the average treatment effect, we derive exact conditional unbiasedness, a within-/between-covariate variance decomposition, the joint population variance minimizer, its separable specialization, singular and constrained cases, and an honest plug-in algorithm. A conditional central limit theorem supports normal-approximation inference when nuisance fitting, coefficient tuning, and final evaluation use independent data blocks; the present theory does not cover cross-fitted sample reuse. For conditional effects, we formulate RCT-valid DR-Fusion and R-Fusion losses, audit their nuisance sensitivity, and develop general two-source learning bounds with linear-sieve and additive-spline specializations. Exact density-ratio transport covers unequal covariate marginals; classifier odds, Riesz learning, and loss-targeted balancing enter an estimated-weight regime with explicit drift. Existing simulations support the ATE construction in several small-trial regimes. CATE experiments and real-outcome analyses remain planned, so neither the theory nor the current evidence establishes universal finite-sample improvement for estimated coefficients.

Contents

1. Introduction	3
1.1. Categorized literature review	3
2. Background and Problem Setting	4
2.1. RCT-OBS problem setting and notation	4
2.2. Existing data-fusion estimators	6

3. Prediction-Powered Fusion for the ATE	7
3.1. Estimator, exact variance, and population oracles	7
3.2. Estimable honest algorithm	10
3.2.1. Coefficient estimation and honest evaluation	10
3.2.2. Estimating the density ratio	12
4. Data Fusion for CATE Estimation	17
4.1. CATE targets and RCT-valid learning losses	17
4.2. Orthogonality of learning equations and nuisance sensitivity	20
4.3. General two-source learning theory	22
4.4. Choosing the fusion coefficients	25
4.5. Linear-sieve and additive-spline specialization	30
4.6. Implementation boundaries and open theory	35
5. Experiments	35
5.1. Completed ATE benchmarks	35
5.2. Planned CATE evaluation and interpretation	37
5.3. Scope, failure modes, and conclusion	37
A. Omitted Preliminaries	41
A.1. Prediction-powered inference and PPI^{++}	41
B. Simulation Details	42
C. Proof Details	44
C.1. Conditional validity of the trial pseudo-outcome	44
C.2. Proof of Corollary 1	44
C.3. Proof of Theorem 1	44
C.4. Proof of Corollary 1.1	45
C.5. Proof of Proposition 2	45
C.6. Proof of Theorem 2	45
C.7. Compact constraints	46
C.8. Proof of Corollary 2.1	46
C.9. Prediction-powered CATE losses and finite-sample theory	46
C.10. Proofs for general two-source learning	46
C.11. Proofs for coefficient selection	47
C.12. Proofs for the orthogonality audit	48
C.13. A two-stratum oracle calculation	51
C.14. Honest variance estimation and evaluation-sample inference	51
C.15. Proof of Theorem 3	51
C.16. Density-ratio characterizations and estimated-ratio inheritance	52
C.17. Counterexamples to adaptive reuse	54

1. Introduction

1.1. Categorized literature review

Prediction-powered inference. Prediction-powered inference uses a large set of machine predictions together with a smaller labeled sample, correcting the prediction-based estimate with labeled residuals so that the statistical target does not require the predictor to be correct (Angelopoulos et al., 2023). This separation between predictive quality and inferential validity is the starting point for the present construction. Randomized observations play the role of causally valid labeled data, while observational outcome models supply potentially useful predictions.

Adaptive prediction-powered inference. PPI⁺⁺ introduces data-adaptive power tuning that adjusts how strongly predictions enter an estimating equation and reports computational and statistical gains over the original construction (Angelopoulos et al., 2024). Its central lesson is that a prediction should be used in proportion to its empirical value rather than being accepted as truth. Our coefficient ω follows this control-variate logic, but the target and score are causal quantities identified by the randomized design.

Causal data fusion and transport. General causal data-fusion theory distinguishes data collected under different regimes and populations and asks whether a target causal query is identified from their combination (Bareinboim and Pearl, 2016). Reviews of causal data integration likewise separate randomized trials, observational studies, historical controls, and heterogeneous target populations (Shi et al., 2023). The present paper assumes that the trial already identifies the target effect and studies an estimation question: how can observational predictions improve precision without replacing randomized identifying information?

Combining randomized and observational estimates. Existing estimators combine experimental and observational effect estimates through shrinkage, bias detection, or adaptive weighting. Shrinkage estimators can reduce risk when observational bias is sufficiently controlled (Rosenman et al., 2023), while adaptive combination methods move toward the randomized estimator when observational bias is detected (Cheng and Cai, 2021). These approaches combine effect estimates from the two sources. Our construction instead inserts the observational outcome model into an RCT-valid pseudo-outcome and separately centers a prediction-based covariate statistic.

CATE learners and orthogonal losses. Meta-learners convert standard regression algorithms into conditional effect estimators (Künzel et al., 2019). The R-learner residualizes outcomes and treatments to isolate treatment-effect heterogeneity (Nie and Wager, 2021), while doubly robust pseudo-outcome regression can retain favorable error structure under nuisance estimation (Kennedy, 2023). Direct multi-source learners have also been proposed for heterogeneous causal data fusion (Li et al., 2022). These results motivate treating the prediction-powered construction as a loss wrapper that can be paired with a linear sieve or a fixed learned representation.

Covariate shift and balancing. When trial and observational covariate marginals differ, exact density-ratio weighting transports fixed covariate functions between populations. Riesz-representer methods provide an optimization perspective on such linear functionals and on debiasing (Chernozhukov et al., 2021). Density-ratio estimation, generic balancing, and Riesz learning are established tools. The project-specific question is whether they can be targeted to the generated CATE loss class $\{(\hat{g} - t)^2 : t \in \mathcal{T}\}$, where $\hat{g}$ is the fitted OBS treatment-effect prediction defined later in Equation (12), while preserving an RCT-defined risk; we treat this specialization as a working

research direction rather than a settled novelty claim.

2. Background and Problem Setting

2.1. RCT–OBS problem setting and notation

Let $\mathcal{K} \triangleq \{\text{eval}, \text{nuis}, \text{tune}\}$ index the three statistical roles assigned to the samples. For each source $S \in \{R, O\}$, where R and O denote the randomized and observational sources, respectively, and each role $k \in \mathcal{K}$, define

$$D_S^k \triangleq \{V_{S,i}^k = (X_{S,i}^k, A_{S,i}^k, Y_{S,i}^k) : i = 1, \dots, n_S^k\} \sim P_S^{\otimes n_S^k}, \quad (1)$$

as the sample from source S reserved for role k . Here X takes values in the covariate space $\mathcal{X}$, $A \in \{0, 1\}$ is treatment, and Y is the observed outcome. For each $S \in \{R, O\}$, the datasets D_S^{eval} , D_S^{nuis} , and D_S^{tune} are used for final evaluation, nuisance and density-ratio fitting, and coefficient or candidate tuning, respectively.

We impose the following independence condition across the six role-specific samples.

Assumption 1 (Honest sample separation). The six datasets $\{D_S^k : S \in \{R, O\}, k \in \mathcal{K}\}$ are mutually independent.

Set $n \triangleq n_R^{\text{eval}}$ and $N \triangleq n_O^{\text{eval}}$, and define $\mathbb{P}_{R,n}h \triangleq n^{-1} \sum_i h(V_{R,i}^{\text{eval}})$ and $\mathbb{P}_{O,N}h \triangleq N^{-1} \sum_j h(V_{O,j}^{\text{eval}})$. Every displayed moment is finite; square-integrability and stronger moment conditions are stated locally when variance estimation or normal-approximation inference requires them.

Let $Y(1)$ and $Y(0)$ denote potential outcomes. The RCT-population conditional and average treatment effects are

$$\tau(x) \triangleq \mathbb{E}_R\{Y(1) - Y(0) \mid X = x\}, \quad (2)$$

$$\theta \triangleq \mathbb{E}_R\{\tau(X)\}. \quad (3)$$

For causal identification in the trial population, we impose the following assumptions under P_R .

Assumption 2 (Trial identification). The following assumptions hold:

1. $Y = Y(A)$ almost surely;
2. $\{Y(0), Y(1)\} \perp A \mid X$ under the interventional distribution P_R ;
3. The trial propensity $e(x) \triangleq P_R(A = 1 \mid X = x)$ is known and satisfies the positivity: $\epsilon < e(X) < 1 - \epsilon$ almost surely for a constant $\epsilon > 0$.

We study both the common-covariate regime $P_R^X = P_O^X$, formalized in Assumption 3, and the more general covariate-shift regime $P_R^X \neq P_O^X$, formalized in Assumption 4.

Definition 1 (Target-to-source density ratio). Whenever $P_R^X \ll P_O^X$, define the target-to-source density ratio by

$$r_0(x) \triangleq \frac{dP_R^X}{dP_O^X}(x).$$

It is the Radon–Nikodym derivative that transports an OBS covariate expectation to the RCT covariate population.

Assumption 3 (Common covariate marginal). The trial and observational covariates have the same marginal law, $P_R^X = P_O^X$. This assumption does not equate the treatment or outcome mechanisms across sources, and the observational treatment contrast may remain confounded. In this regime, Definition 1 gives $r_0(x) = 1$ for P_O^X -almost every x .

Assumption 4 (Covariate shift and support overlap). Under covariate shift, $P_R^X \ll P_O^X$, so the RCT covariate population has no support outside the OBS covariate population. Definition 1 therefore defines r_0 . This assumption establishes existence of the ratio; it does not assert that r_0 is known or exactly estimated.

Assumptions 3 and 4 describe alternative covariate regimes, not cumulative restrictions. Under Assumption 3, the exact density ratio is $r_0 = 1$, so no transport weighting is needed. When the covariate marginals differ, Assumption 4 is needed to convert OBS covariate expectations into expectations under the RCT population.

For $a \in \{0, 1\}$, define the RCT outcome-regression target $\mu_{R,a} : \mathcal{X} \rightarrow \mathbb{R}$ by

$$\mu_{R,a}(x) \triangleq \mathbb{E}_R[Y \mid X = x, A = a].$$

For each $a \in \{0, 1\}$, let $\hat{\mu}_{R,a} : \mathcal{X} \rightarrow \mathbb{R}$ denote a fixed pre-evaluation predictor of $\mu_{R,a}$ trained only on D_R^{nuis} . Define

$$\hat{\mu}_R \triangleq (\hat{\mu}_{R,0}, \hat{\mu}_{R,1}). \quad (4)$$

We define the following AIPW score.

Definition 2 (RCT-valid AIPW score). Let $q \triangleq (q_0, q_1)$ be a fixed pair of measurable functions from $\mathcal{X}$ to $\mathbb{R}$. Define

$$\varphi(V; q) \triangleq q_1(X) - q_0(X) + \frac{A}{e(X)}\{Y - q_1(X)\} - \frac{1 - A}{1 - e(X)}\{Y - q_0(X)\}. \quad (5)$$

This score gives a valid causal estimate:

Proposition 1 (RCT-validity for an arbitrary outcome regression). Suppose Assumption 2 holds. Let q be a measurable function pair fixed independently of the RCT evaluation sample, and suppose $\varphi(V; q)$ is integrable. Then

$$\mathbb{E}_R\{\varphi(V; q) \mid X\} = \tau(X). \quad (6)$$

Definition 3 (RCT-only AIPW estimator). Define the RCT-only AIPW estimator on the evaluation sample by

$$\hat{\theta}_{\text{AIPW},R} \triangleq \mathbb{P}_{R,n}\varphi(V; \hat{\mu}_R). \quad (7)$$

Corollary 1 (RCT-only AIPW recovery). Suppose Assumptions 2 and 1 hold and $\varphi(V; \hat{\mu}_R)$ is integrable. Conditional on the nuisance data used to fit $\hat{\mu}_R$, $\mathbb{E}\{\hat{\theta}_{\text{AIPW},R} \mid D_R^{\text{nuis}}\} = \theta$.

For $a \in \{0, 1\}$, define the OBS outcome-regression target $\mu_{O,a} : \mathcal{X} \rightarrow \mathbb{R}$ by

$$\mu_{O,a}(x) \triangleq \mathbb{E}_O[Y \mid X = x, A = a].$$

For each $a \in \{0, 1\}$, let $\hat{\mu}_{O,a} : \mathcal{X} \rightarrow \mathbb{R}$ denote a fixed pre-evaluation predictor of $\mu_{O,a}$ trained only on D_O^{nuis} . Define

$$\hat{\mu}_O \triangleq (\hat{\mu}_{O,0}, \hat{\mu}_{O,1}). \quad (8)$$

The RCT-validity result also permits the RCT and OBS outcome regressions to be combined without changing the conditional causal target.

Definition 4 (Fusion score). For $\lambda \in \mathbb{R}$, define

$$\hat{\mu}_\lambda \triangleq (1 - \lambda)\hat{\mu}_R + \lambda\hat{\mu}_O, \quad (9)$$

and the fused score as

$$Z_\lambda \triangleq \varphi(V; \hat{\mu}_\lambda). \quad (10)$$

From $Z_0 = \varphi(V; \hat{\mu}_R)$, $Z_1 = \varphi(V; \hat{\mu}_O)$ and $\Delta \triangleq Z_1 - Z_0$, we can rewrite the fused score as

$$Z_\lambda = Z_0 + \lambda\Delta. \quad (11)$$

2.2. Existing data-fusion estimators

Data integration first asks whether information from different regimes or populations identifies a common target. Transport and general data-fusion frameworks make the source and target populations explicit (Bareinboim and Pearl, 2016; Shi et al., 2023; Colnet et al., 2020). Our target is already identified by the RCT; transport is used to align fixed OBS covariate functions with the RCT population.

Robust borrowing methods instead trade potential efficiency gains against observational bias. They include shrinkage, compatibility testing, and adaptive weighting that returns toward the randomized estimate when the sources disagree (Rosenman et al., 2023; Cheng and Cai, 2021). These methods clarify why observational information can help without implying a universal gain.

For heterogeneous effects, direct multi-source learners combine weighted pseudo-outcomes, while experimental-grounding and RCT-OBS procedures use the trial to constrain or diagnose observational HTE estimates (Li et al., 2022; Kallus et al., 2018). The elastic procedure of Yang et al. (2023) tests compatibility before pooling real-world data for prespecified effect modifiers. The confounding-function approach of Yang et al. (2025) jointly models an HTE function and an observational confounding function under explicit identification conditions.

Our construction differs at the fusion object. The coefficient λ changes the outcome regression inside an RCT-valid score, whereas the second coefficient scales a transported mean-zero control variate built from a fitted OBS treatment-effect prediction. This PPI-style use of prediction does not combine two causal effect estimates and does not require the OBS contrast to equal the CATE.

3. Prediction-Powered Fusion for the ATE

Throughout this section, Assumptions 2 and 1 hold, the evaluation sample sizes satisfy $n, N > 0$, and either Assumption 3 or Assumption 4 holds. Thus, the exact density ratio r_0 in Definition 1 exists and equals one under the common-covariate regime. A feasible estimator may use a fixed candidate weight r ; results that require the exact weight state $r = r_0$ explicitly.

3.1. Estimator, exact variance, and population oracles

Definition 5 (ATE fusion estimator and its components). Using the fitted OBS outcome regression $\hat{\mu}_O$ from Equation (8), define the fitted observational treatment-effect prediction by

$$\hat{g}(x) \triangleq \hat{\mu}_{O,1}(x) - \hat{\mu}_{O,0}(x). \quad (12)$$

Let $r : \mathcal{X} \rightarrow [0, \infty)$ be a fixed measurable candidate transport weight. For fixed (λ, ω) , define the ATE fusion estimator by

$$\hat{\theta}_r(\lambda, \omega) \triangleq \mathbb{P}_{R,n} Z_\lambda + \omega \{ \mathbb{P}_{O,N} \{ r(X) \hat{g}(X) \} - \mathbb{P}_{R,n} \hat{g}(X) \}, \quad (13)$$

where Z_λ is the RCT-valid score in Equation (10). The fitted outcome regressions $\hat{\mu}_R$ and $\hat{\mu}_O$ are defined in Equations (4) and (8), respectively. The coefficient λ blends these regressions inside Z_λ , whereas ω scales the transported control variate. Exact transport uses $r = r_0$.

At the coefficient origin,

$$\hat{\theta}_r(0, 0) = \mathbb{P}_{R,n} Z_0 = \hat{\theta}_{\text{AIPW},R},$$

where $\hat{\theta}_{\text{AIPW},R}$ is defined in Equation (7), so the family contains the RCT-only AIPW estimator.

The estimator depends on three learned components: the outcome-regression blend inside $Z_\lambda \triangleq \varphi(V; \hat{\mu}_\lambda)$, the OBS prediction $\hat{g}$, and the density ratio r . The next result states exactly which components can change its population mean.

Theorem 1 (Exact target, density-ratio drift, and variance). For every fixed (λ, ω) , arbitrary fixed outcome-regression fits and a fixed OBS prediction satisfy, whenever the displayed expectations are finite,

$$\mathbb{E}\{\hat{\theta}_{r_0}(\lambda, \omega)\} = \theta. \quad (14)$$

For any fixed candidate weight $\tilde{r}$, its exact mean drift is

$$\mathbb{E}\{\hat{\theta}_{\tilde{r}}(\lambda, \omega)\} - \theta = \omega \mathbb{E}_O [\{\tilde{r}(X) - r_0(X)\} \hat{g}(X)]. \quad (15)$$

If $Z_\lambda - \omega \hat{g}(X)$ and $r_0(X) \hat{g}(X)$ are square-integrable, then

$$\text{Var}\{\hat{\theta}_{r_0}(\lambda, \omega)\} = \frac{1}{n} \text{Var}_R\{Z_\lambda - \omega \hat{g}(X)\} + \frac{\omega^2}{N} \text{Var}_O\{r_0(X) \hat{g}(X)\}. \quad (16)$$

Equation (14) shows that, under exact transport $r = r_0$, the expectation of $\hat{\theta}_{r_0}(\lambda, \omega)$ is unchanged by the fixed outcome regressions $\hat{\mu}_R$ and $\hat{\mu}_O$ or by the fixed OBS prediction $\hat{g}$. If a candidate weight $\tilde{r}$ differs from r_0 , Equation (15) gives the resulting mean error exactly.

We next minimize the exact variance in Equation (16) over the two fusion coefficients.

Definition 6 (ATE variance coefficients). Recall from Equation (10) and Section 2.1 that $Z_0 \triangleq \varphi(V; \hat{\mu}_R)$ and $\Delta \triangleq Z_1 - Z_0$; $\hat{g}$ is the OBS prediction in Equation (12). Define

$$\begin{aligned} A &\triangleq \text{Var}_R(\Delta), & B &\triangleq \text{Var}_R(\hat{g}) + \frac{n}{N} \text{Var}_O(r_0 \hat{g}), \\ C &\triangleq \text{Cov}_R(Z_0, \Delta), & D &\triangleq \text{Cov}_R(Z_0, \hat{g}). \end{aligned} \quad (17)$$

Here A and B are the quadratic costs of changing λ and ω , while C and D determine their linear slopes. Also define

$$\mathbf{H}_{\text{ATE}} \triangleq \begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix}, \quad \boldsymbol{\ell} \triangleq \begin{pmatrix} C \\ -D \end{pmatrix}, \quad \boldsymbol{\beta} \triangleq \begin{pmatrix} \lambda \\ \omega \end{pmatrix}.$$

The following corollary rewrites the variance in Equation (16) using the four coefficients A , B , C , and D .

Corollary 1.1 (Separated ATE variance quadratic). If the displayed scores have finite second moments, then

$$\text{Var}\{\hat{\theta}_{r_0}(\lambda, \omega)\} = \frac{1}{n} \left\{ \text{Var}_R(Z_0) + 2C\lambda - 2D\omega + A\lambda^2 + B\omega^2 \right\}. \quad (18)$$

Removing the constant term from Equation (18) gives the following equivalent optimization objective.

Proposition 2 (Equivalent ATE coefficient objective). Define

$$\mathcal{J}_{\text{ATE}}(\lambda, \omega) \triangleq 2C\lambda - 2D\omega + A\lambda^2 + B\omega^2.$$

For every nonempty compact $\Theta \subseteq \mathbb{R}^2$,

$$\arg \min_{(\lambda, \omega) \in \Theta} \text{Var}\{\widehat{\theta}_{r_0}(\lambda, \omega)\} = \arg \min_{(\lambda, \omega) \in \Theta} \mathcal{J}_{\text{ATE}}(\lambda, \omega).$$

The next theorem gives the closed-form variance-minimizing coefficients and their ridge-regularized counterparts.

Theorem 2 (Population and ridge oracles). Suppose the displayed scores have finite second moments and $A, B > 0$. For every $\rho \geq 0$, the unique ridge optimizer is

$$\begin{pmatrix} \lambda_\rho^* \\ \omega_\rho^* \end{pmatrix} \triangleq \arg \min_{(\lambda, \omega) \in \mathbb{R}^2} \left\{ \mathcal{J}_{\text{ATE}}(\lambda, \omega) + \rho(\lambda^2 + \omega^2) \right\} = \begin{pmatrix} -C/(A + \rho) \\ D/(B + \rho) \end{pmatrix}. \quad (19)$$

At $\rho = 0$, this is the unique minimizer of the exact population variance:

$$\lambda^* \triangleq \lambda_0^* = -\frac{C}{A}, \quad \omega^* \triangleq \omega_0^* = \frac{D}{B}. \quad (20)$$

Corollary 2.1 (Minimum population variance). Suppose the displayed scores have finite second moments and $A, B > 0$. Define

$$\mathcal{V}^* \triangleq \min_{(\lambda, \omega) \in \mathbb{R}^2} \text{Var}\{\widehat{\theta}_{r_0}(\lambda, \omega)\}. \quad (21)$$

Let $\mathcal{V}_0 \triangleq \text{Var}_R(Z_0)/n$. Then

$$\mathcal{V}^* = \mathcal{V}_0 - \frac{1}{n} \left(\frac{C^2}{A} + \frac{D^2}{B} \right), \quad \mathcal{V}_0 - \mathcal{V}^* = \frac{1}{n} \left(\frac{C^2}{A} + \frac{D^2}{B} \right) \geq 0. \quad (22)$$

The next theorem separates the two sources of this oracle variance reduction.

Theorem 3 (Exact-ratio oracle variance comparison). Suppose the scores in Equation (16) have finite second moments and $A, B > 0$. Define $\tilde{\mathcal{V}}(\lambda, \omega) \triangleq n \text{Var}\{\widehat{\theta}_{r_0}(\lambda, \omega)\}$ and $\sigma_0^2 \triangleq \text{Var}_R(Z_0)$. Here “PPI⁺⁺-only” means the member of our family that fixes $\lambda = 0$ and optimally selects ω . Then

$$\begin{aligned} \tilde{\mathcal{V}}_{\text{RCT}} &\triangleq \tilde{\mathcal{V}}(0, 0) = \sigma_0^2, \\ \tilde{\mathcal{V}}_{\text{PPI}^{++}} &\triangleq \tilde{\mathcal{V}}(0, D/B) = \sigma_0^2 - \frac{D^2}{B}, \\ \tilde{\mathcal{V}}_{\text{joint}} &\triangleq \tilde{\mathcal{V}}(-C/A, D/B) = \sigma_0^2 - \frac{D^2}{B} - \frac{C^2}{A}. \end{aligned} \quad (23)$$

Therefore $\tilde{\mathcal{V}}_{\text{joint}} \leq \tilde{\mathcal{V}}_{\text{PPI}^{++}} \leq \tilde{\mathcal{V}}_{\text{RCT}}$.

Thus, with population-oracle coefficients and the exact density ratio, the joint ATE estimator has variance no larger than either the PPI⁺⁺-only estimator or the RCT-only estimator.

3.2. Estimable honest algorithm

This subsection gives a finite-sample algorithm for estimating the two fusion coefficients.

We first define the feasible region over which λ and ω are selected.

Definition 7 (Coefficient rectangle and interval projection). Define the nonempty compact coefficient rectangle by

$$\Theta_{\text{rect}} \triangleq [L_\lambda, U_\lambda] \times [L_\omega, U_\omega]. \quad (24)$$

For a closed interval $[L, U]$ with $L \leq U$, define

$$\text{Proj}_{[L, U]}(u) \triangleq \min\{U, \max(L, u)\}. \quad (25)$$

This operator returns u when it lies in the interval and otherwise returns the nearest endpoint.

For positive A, B , the separated objective gives the coordinatewise projection

$$\lambda_{\Theta_{\text{rect}}}^* = \text{Proj}_{[L_\lambda, U_\lambda]}(-C/A), \quad \omega_{\Theta_{\text{rect}}}^* = \text{Proj}_{[L_\omega, U_\omega]}(D/B).$$

The population oracle weakly improves the variance objective whenever $(0, 0)$ is feasible. Estimated coefficients have no universal finite-sample no-harm guarantee.

3.2.1. Coefficient estimation and honest evaluation

For $S \in \{R, O\}$, suppose $n_S^{\text{tune}} \geq 2$. Conditional on the fixed nuisance fits, $\widehat{\text{Cov}}_S$ and $\widehat{\text{Var}}_S$ denote the usual unbiased tuning-sample covariance and variance with denominator $n_S^{\text{tune}} - 1$. The factor n/N below uses the evaluation-sample sizes whose variance is optimized.

Let $\hat{r}$ denote a density-ratio estimator fitted only on the nuisance samples. Under the common-covariate regime, set $\hat{r} \triangleq 1$.

Using the tuning samples, compute the following four empirical analogues of A , B , C , and D in Definition 6:

$$\begin{aligned} \hat{A} &\triangleq \widehat{\text{Var}}_R(\Delta), & \hat{B} &\triangleq \widehat{\text{Var}}_R(\hat{g}) + \frac{n}{N} \widehat{\text{Var}}_O(\hat{r}\hat{g}), \\ \hat{C} &\triangleq \widehat{\text{Cov}}_R(Z_0, \Delta), & \hat{D} &\triangleq \widehat{\text{Cov}}_R(Z_0, \hat{g}). \end{aligned} \quad (26)$$

These four moments define the separated empirical objective

$$\hat{\mathcal{J}}_{\text{ATE}}(\lambda, \omega) \triangleq 2\hat{C}\lambda - 2\hat{D}\omega + \hat{A}\lambda^2 + \hat{B}\omega^2. \quad (27)$$

Before observing the tuning outcomes, prespecify the bounds defining the nonempty compact rectangle Θ_{rect} , require it to contain the origin, and choose positive thresholds t_A and t_B .

Recall that $\hat{\mu}_R = (\hat{\mu}_{R,0}, \hat{\mu}_{R,1})$ and $\hat{\mu}_O = (\hat{\mu}_{O,0}, \hat{\mu}_{O,1})$ are outcome-regression predictors fitted on the RCT and OBS nuisance samples, respectively, and $\hat{g}$ is their fitted OBS contrast from Equation (12). Algorithm 1 states the independent-sample implementation covered by the exact finite-sample results. The nuisance, tuning, and evaluation samples may be supplied directly or obtained by an honest prespecified split.

Algorithm 1: ATE estimation with honest evaluation

Input: Nuisance samples $D_R^{\text{nuis}}, D_O^{\text{nuis}}$, tuning samples $D_R^{\text{tune}}, D_O^{\text{tune}}$, evaluation samples $D_R^{\text{eval}}, D_O^{\text{eval}}$, the rectangle Θ_{rect} , and denominator thresholds.

- 1 Fit $\hat{\mu}_R, \hat{\mu}_O, \hat{g}$, and a density-ratio estimator $\hat{r}$ on $D_R^{\text{nuis}}, D_O^{\text{nuis}}$. The function $\hat{r}$ estimates the exact density ratio r_0 . When $\hat{r} \neq r_0$, Equation (15) gives the resulting mean error.
- 2 Compute the four moments in Equation (26) on $D_R^{\text{tune}}, D_O^{\text{tune}}$, and select the coefficients using the threshold-and-projection rule in Equation (28).
- 3 Set $r = \hat{r}$ in Equation (13), and evaluate it on $D_R^{\text{eval}}, D_O^{\text{eval}}$. Set $\psi_R \triangleq Z_{\hat{\lambda}} - \hat{\omega}\hat{g}$ and $\psi_O \triangleq \hat{\omega}\hat{r}\hat{g}$. Compute $\hat{V} \triangleq \widehat{\text{Var}}_{R,\text{eval}}(\psi_R)/n + \widehat{\text{Var}}_{O,\text{eval}}(\psi_O)/N$.

Output: $\hat{\theta}_{\hat{r}}$ and $\hat{V}$.

How Algorithm 1 selects the coefficients. Because Equation (27) is separated in λ and ω , Algorithm 1 selects the two coefficients coordinatewise and projects each one onto its prespecified interval:

$$\hat{\lambda} \triangleq \begin{cases} \text{Proj}_{[L_{\lambda}, U_{\lambda}]}(-\hat{C}/\hat{A}), & \hat{A} > t_A, \\ 0, & \hat{A} \leq t_A, \end{cases} \quad \hat{\omega} \triangleq \begin{cases} \text{Proj}_{[L_{\omega}, U_{\omega}]}(\hat{D}/\hat{B}), & \hat{B} > t_B, \\ 0, & \hat{B} \leq t_B. \end{cases} \quad (28)$$

Analysis. The four-moment rule in Equation (28) directly estimates the two population-oracle coordinates in Equation (20). The thresholds avoid division by nearly zero empirical curvature, and the projections enforce the prespecified feasible rectangle.

For Algorithm 1, set $r = \hat{r}$ in the generic evaluation scores below. For explicit evaluation notation, define

$$\psi_{R,i} \triangleq Z_{\hat{\lambda}}(V_{R,i}^{\text{eval}}) - \hat{\omega}\hat{g}(X_{R,i}^{\text{eval}}), \quad \psi_{O,j} \triangleq \hat{\omega}r(X_{O,j}^{\text{eval}})\hat{g}(X_{O,j}^{\text{eval}}),$$

Let $\widehat{\text{Var}}_{R,\text{eval}}$ and $\widehat{\text{Var}}_{O,\text{eval}}$ be the usual unbiased sample variances with denominators $n - 1$ and $N - 1$, respectively. Define the fixed-selected-coefficient variance estimator by

$$\hat{V} \triangleq \frac{\widehat{\text{Var}}_{R,\text{eval}}(\psi_R)}{n} + \frac{\widehat{\text{Var}}_{O,\text{eval}}(\psi_O)}{N}. \quad (29)$$

Proposition 3 (Exact variance estimation after honest selection). Let $\hat{\theta}_r$ and $\hat{V}$ be the estimator and variance estimator from Algorithm 1, specialized to $r = \hat{r} = r_0$. Suppose Assumption 1 holds, $n, N \geq 2$, and $\psi_R \in L_2(P_R)$, $\psi_O \in L_2(P_O)$. Conditional on all fitted and selected objects,

$$\mathbb{E}(\hat{\theta}_r) = \theta, \quad \mathbb{E}(\hat{V}) = \text{Var}\{\hat{\theta}_r(\hat{\lambda}, \hat{\omega})\}.$$

When the corresponding unconditional moments exist, the same identities also hold unconditionally because the conditional mean is the nonrandom target θ .

Theorem 4 (Normal approximation after honest evaluation). Let $\hat{\theta}_r$ and $\hat{V}$ be the estimator and variance estimator from Algorithm 1, specialized to $r = \hat{r} = r_0$. Suppose Assumption 1 holds, $n, N \rightarrow \infty$, $n/N \rightarrow \kappa \in [0, \infty)$, and their centered scores have uniformly bounded conditional $(2 + \delta)$ -moments for some $\delta > 0$.

Define the conditional evaluation variance $V_n \triangleq \text{Var}\{\hat{\theta}_r \mid \text{fitted and selected objects}\}$ and assume $nV_n \rightarrow_p \sigma^2 \in (0, \infty)$ and $\hat{V}/V_n \rightarrow_p 1$. Then

$$\frac{\hat{\theta}_r - \theta}{\sqrt{\hat{V}}} \rightsquigarrow N(0, 1),$$

and

$$\hat{\theta}_r \pm z_{1-\alpha/2} \sqrt{\hat{V}}$$

has asymptotic coverage $1 - \alpha$, where $z_{1-\alpha/2}$ is the standard normal quantile. Variance unbiasedness alone does not imply this coverage.

Proposition 3 and Theorem 4 are the exact-ratio benchmarks. The next subsection develops feasible inference when Algorithm 1 uses an estimated ratio $\hat{r}$ that need not equal r_0 .

3.2.2. Estimating the density ratio

This subsection replaces the fixed candidate r above by a fitted ratio $\hat{r}$ in three steps. Propositions 4 and 5 characterize the target r_0 in two ways, as the unique weight that transports OBS averages to RCT averages and as the unique minimizer of a quadratic risk. Definitions 8 to 10 turn these two characterizations into estimators. Proposition 7, Theorem 5, and Corollary 5.1 then track how the error $\hat{r} - r_0$ enters Algorithm 1: it biases the point estimate by an explicit drift but leaves the variance estimator unbiased, and when the drift is small relative to the standard error, the exact-ratio results of Theorems 4 and 3 carry over.

Proposition 4 (Transport characterization). Under either Assumption 3 or Assumption 4, every $h \in L_1(P_R^X)$ satisfies

$$\mathbb{E}_O\{r_0(X)h(X)\} = \mathbb{E}_R\{h(X)\}. \quad (30)$$

Conversely, if $\tilde{r} \geq 0$, $\mathbb{E}_O(\tilde{r}) = 1$, and Equation (30) holds with $\tilde{r}$ in place of r_0 for every bounded measurable h , then $\tilde{r} = r_0$ P_O^X -almost surely.

Proposition 5 (Riesz-risk characterization). If $r_0 \in L_2(P_O^X)$, then

$$r_0 \in \arg \min_{a \in L_2(P_O^X)} E_O\{a(X)^2\} - 2\mathbb{E}_R\{a(X)\}, \quad (31)$$

The minimizer is unique up to P_O^X -almost-sure equality.

Eq. (31) motivates the following learner.

Definition 8 (Empirical Riesz ratio learner). For $S \in \{R, O\}$, let $\mathbb{P}_S^{\text{nuis}}$ denote the empirical average over D_S^{nuis} . For a prespecified function class $\mathcal{A}$, penalty Pen , and $\rho \geq 0$, use a prespecified deterministic rule to break ties among minimizers. Whenever the displayed argmin set is nonempty, choose

$$\hat{r}_{\text{RZ}} \in \arg \min_{a \in \mathcal{A}} \left[\mathbb{P}_O^{\text{nuis}}(a^2) - 2\mathbb{P}_R^{\text{nuis}}(a) + \rho \text{Pen}(a) \right]. \quad (32)$$

The classifier and Riesz routes are alternative base learners; we do not average them.

Equation (32) is Riesz-representer learning of the type used in Auto-DML (Chernozhukov et al., 2021, 2022).

The next lemma specializes Definition 8 to a linear-ridge class and gives explicit conditions for a unique solution.

Lemma 1 (Linear Riesz solution). Let $a_\gamma(x) \triangleq \phi(x)^\top \gamma$, where $\phi : \mathcal{X} \rightarrow \mathbb{R}^p$. Let $\text{Pen}(a_\gamma) \triangleq \|\gamma\|_2^2$. The empirical Gram matrix, namely the average of the feature outer products over the OBS nuisance sample, is $\mathbb{P}_O^{\text{nuis}}\{\phi(X)\phi(X)^\top\}$. If $\rho > 0$, or if this matrix is nonsingular when $\rho = 0$, then

$$\hat{\gamma}_\rho \triangleq \left[\mathbb{P}_O^{\text{nuis}}\{\phi(X)\phi(X)^\top\} + \rho I_p \right]^{-1} \mathbb{P}_R^{\text{nuis}}\{\phi(X)\}. \quad (33)$$

The classifier learns the ratio indirectly through source prediction.

Definition 9 (Classifier density-ratio estimator). Pool the nuisance-sample covariates and label their source by $S = 1$ for RCT and $S = 0$ for OBS. Define

$$\pi \triangleq P(S = 1), \quad s_0(x) \triangleq P(S = 1 \mid X = x).$$

When $0 < \pi < 1$, Bayes' rule gives

$$r_0(x) = \frac{1 - \pi}{\pi} \frac{s_0(x)}{1 - s_0(x)}. \quad (34)$$

For a fitted domain probability $\hat{s}$, let $\hat{\pi}$ denote the effective RCT proportion in classifier

training and define

$$\hat{r}_{\text{CLS}}(x) \triangleq \frac{1 - \hat{\pi}}{\hat{\pi}} \frac{\hat{s}(x)}{1 - \hat{s}(x)}. \quad (35)$$

For an unweighted pooled classifier, $\hat{\pi} \triangleq n_R^{\text{nuis}} / (n_R^{\text{nuis}} + n_O^{\text{nuis}})$. Class weighting or resampling changes the effective training prior, so the prior-odds factor must be corrected accordingly. The odds require $0 < \hat{s}(x) < 1$ where the ratio is used. Probability clipping can stabilize the estimate, but it changes the exact density-ratio target.

The Riesz learner in Definition 8 and the classifier in Definition 9 are the two primary ratio-estimation routes. After either route has been fitted, one may optionally calibrate its positive ratio to selected ATE features.

Definition 10 (Calibrated automatic-balancing ratio). Let $\hat{r}_{\text{base}} > 0$ come from either the classifier or Riesz route. Let $\phi : \mathcal{X} \rightarrow \mathbb{R}^p$ be prespecified with no intercept. Whenever the displayed argmin set is nonempty, choose a deterministically tie-broken minimizer

$$\hat{\xi} \in \arg \min_{\xi \in \mathbb{R}^p} \left[\log \mathbb{P}_O^{\text{nuis}} \{ \hat{r}_{\text{base}}(X) e^{\xi^\top \phi(X)} \} - \xi^\top \mathbb{P}_R^{\text{nuis}} \{ \phi(X) \} \right], \quad (36)$$

$$\hat{r}_{\text{BAL}}(x) \triangleq \frac{\hat{r}_{\text{base}}(x) e^{\hat{\xi}^\top \phi(x)}}{\mathbb{P}_O^{\text{nuis}} \{ \hat{r}_{\text{base}}(X) e^{\hat{\xi}^\top \phi(X)} \}}. \quad (37)$$

Normalization already balances the constant, so an intercept would be nonidentified. The minimal ATE feature is $\phi_{\text{ATE}}(X) \triangleq \hat{g}(X)$; a richer finite map can add $\hat{g}^2$, fitted regressions, or a small spline or frozen-neural basis. This exponential tilt follows the calibration principle behind entropy balancing (Hainmueller, 2012).

Proposition 6 (Automatic balance). If the objective in Definition 10 has an interior minimizer, then $\hat{r}_{\text{BAL}} > 0$, $\mathbb{P}_O^{\text{nuis}}(\hat{r}_{\text{BAL}}) = 1$, and

$$\mathbb{P}_O^{\text{nuis}} \{ \hat{r}_{\text{BAL}}(X) \phi(X) \} = \mathbb{P}_R^{\text{nuis}} \{ \phi(X) \}. \quad (38)$$

These are p finite moment equations, not pointwise balance. They are analogous to covariate-balancing propensity scores, which concern treatment rather than source assignment (Imai and Ratkovic, 2014). Exact feasibility requires the RCT feature mean to lie in the relative interior of the convex hull of the positively weighted OBS feature vectors; otherwise a prespecified approximate-balance rule is required.

We now quantify how the error $\hat{r} - r_0$ propagates through Algorithm 1. The estimated ratio enters the algorithm at exactly two places. In the tuning step it enters the moment $\hat{B}$ of Equation (26), which sets $\hat{\omega}$ through Equation (28). In the evaluation step it enters the OBS score $\psi_O = \hat{\omega} \hat{r} \hat{g}$, which feeds both $\hat{\theta}_{\hat{r}}$ and $\hat{V}$ in Equation (29). The coefficient $\hat{\lambda}$ uses only $\hat{A}$ and $\hat{C}$, so it does not depend on $\hat{r}$. The next proposition treats the two places in turn and then records what they imply for the bias and the variance of $\hat{\theta}_{\hat{r}}$.

Proposition 7 (Effect of the estimated ratio on Algorithm 1). Let $\widehat{B}_{\widehat{r}}$ denote $\widehat{B}$ in Equation (26) evaluated with the estimated ratio $\widehat{r}$, and let $\widehat{B}_{r_0}$ denote the same expression evaluated with the oracle ratio r_0 . Let $\widehat{\theta}_{\widehat{r}}$ and $\widehat{\theta}_{r_0}$ be the estimator in Equation (13) with the same selected coefficients $(\widehat{\lambda}, \widehat{\omega})$ and with $r = \widehat{r}$ and $r = r_0$, respectively. Suppose Assumption 1 holds, $n, N \geq 2$, $\psi_R \in L_2(P_R)$, $\psi_O \in L_2(P_O)$, $\widehat{g} \in L_1(P_R)$, and $r_0\widehat{g}, \widehat{r}\widehat{g} \in L_1(P_O)$.

1. (Tuning step.) On the tuning samples,

$$\widehat{B}_{\widehat{r}} - \widehat{B}_{r_0} = \frac{n}{N} \left[2\widehat{\text{Cov}}_O\{r_0\widehat{g}, (\widehat{r} - r_0)\widehat{g}\} + \widehat{\text{Var}}_O\{(\widehat{r} - r_0)\widehat{g}\} \right]. \quad (39)$$

Hence the ratio error changes $\widehat{\omega} = \widehat{D}/\widehat{B}$ only through its denominator and the threshold t_B in Equation (28), and it leaves $\widehat{\lambda}$ unchanged.

2. (Evaluation step.) On the evaluation samples,

$$\widehat{\theta}_{\widehat{r}} - \widehat{\theta}_{r_0} = \widehat{\omega} \mathbb{P}_{O,N}\{(\widehat{r} - r_0)\widehat{g}\}. \quad (40)$$

3. (Bias and variance.) Conditional on all fitted and selected objects,

$$\mathbb{E}(\widehat{\theta}_{\widehat{r}}) = \theta + \widehat{\omega}\{\mathbb{E}_O(\widehat{r}\widehat{g}) - \mathbb{E}_R(\widehat{g})\}, \quad \mathbb{E}(\widehat{V}) = \text{Var}(\widehat{\theta}_{\widehat{r}}). \quad (41)$$

Item 1 is the coefficient-selection path and item 2 is the evaluation path. Both are small when the OBS sample is much larger than the trial: the difference in item 1 carries the factor n/N , and the difference in item 2 has conditional variance $\widehat{\omega}^2 \text{Var}_O\{(\widehat{r} - r_0)\widehat{g}\}/N$. Item 3 says what remains. The ratio error biases the point estimate by the drift $\widehat{\omega}\{\mathbb{E}_O(\widehat{r}\widehat{g}) - \mathbb{E}_R(\widehat{g})\}$, which is $\widehat{\omega}$ times the transport error of $\widehat{r}$ on the single function $\widehat{g}$, while $\widehat{V}$ stays an unbiased estimator of the conditional variance because it is built from the same scores as $\widehat{\theta}_{\widehat{r}}$. The drift vanishes when $\mathbb{E}_O(\widehat{r}\widehat{g}) = \mathbb{E}_R(\widehat{g})$, which is the population version of the balance that Definition 10 enforces on the nuisance sample with $\phi = \widehat{g}$, and it also vanishes when $\widehat{\omega} = 0$. When the corresponding unconditional moments exist, the unconditional mean of $\widehat{\theta}_{\widehat{r}}$ is θ plus the expected drift, and the unconditional variance additionally contains the between-fit variation of the drift, which $\widehat{V}$ does not capture. Therefore the only new requirement for Wald inference with $\widehat{r}$ is that the drift be small relative to the standard error.

Theorem 5 (Feasible normal approximation with an estimated ratio). Let $\widehat{\theta}_{\widehat{r}}$ and $\widehat{V}$ be the outputs of Algorithm 1 with the estimated ratio $\widehat{r}$, and let $V_n \triangleq \text{Var}(\widehat{\theta}_{\widehat{r}} \mid \text{fitted and selected objects})$ be the conditional variance in item 3 of Proposition 7. Suppose Assumption 1 holds, $n, N \rightarrow \infty$, $n/N \rightarrow \kappa \in [0, \infty)$, and, for some $\delta > 0$, the centered evaluation scores have conditional $(2+\delta)$ -moments that are uniformly bounded in probability. Suppose also that

$$nV_n \rightarrow_p \sigma^2 \in (0, \infty), \quad \frac{\widehat{\omega}\{\mathbb{E}_O(\widehat{r}\widehat{g}) - \mathbb{E}_R(\widehat{g})\}}{\sqrt{V_n}} = o_p(1). \quad (42)$$

Then

$$\frac{\widehat{V}}{V_n} \rightarrow_p 1, \quad \frac{\widehat{\theta}_{\widehat{r}} - \theta}{\sqrt{\widehat{V}}} \rightsquigarrow N(0, 1), \quad (43)$$

and the interval

$$\hat{\theta}_{\hat{r}} \pm z_{1-\alpha/2} \sqrt{\hat{V}} \quad (44)$$

has asymptotic coverage $1 - \alpha$, where $z_{1-\alpha/2}$ is the standard normal quantile.

Equation (42) states two requirements. The first is that the conditional variance shrinks at the parametric rate with a nondegenerate limit, so that the standard error $\sqrt{V_n}$ is of order $n^{-1/2}$. The second is that the bias caused by the ratio error, the drift of Proposition 7, is negligible relative to that standard error. Equation (43) gives the two conclusions that a Wald interval needs: the variance estimator is consistent for the conditional variance, and the studentized estimator is asymptotically standard normal. Theorem 5 therefore permits $\hat{r} \neq r_0$, and the price is the drift condition. If the drift is not negligible, the interval needs further justification, which may use influence functions, direct bias control, or a different orthogonal construction that accounts for first-stage variation.

Corollary 5.1 (Inheritance of the exact-ratio oracle ordering). Index the three estimators of Theorem 3, built from the same fitted objects $Z_0, \Delta, \hat{g}, \hat{r}$, by $k \in \{\text{RCT}, \text{PPI}^{++}, \text{joint}\}$. Their population-oracle coefficients are

$$\begin{aligned} (\lambda_{\text{RCT}}^*, \omega_{\text{RCT}}^*) &\triangleq (0, 0), & (\lambda_{\text{PPI}^{++}}^*, \omega_{\text{PPI}^{++}}^*) &\triangleq (0, D/B), \\ (\lambda_{\text{joint}}^*, \omega_{\text{joint}}^*) &\triangleq (-C/A, D/B), \end{aligned}$$

and their feasible coefficients $(\hat{\lambda}_k, \hat{\omega}_k)$ are the entries of Equation (28), with $\hat{\lambda}_k \triangleq 0$ for $k \neq \text{joint}$ and $\hat{\omega}_{\text{RCT}} \triangleq 0$. Write $\hat{\theta}_{\hat{r},k} \triangleq \hat{\theta}_{\hat{r}}(\hat{\lambda}_k, \hat{\omega}_k)$, and let σ_k^2 denote the limit σ^2 in Equation (42) for estimator k . Suppose that, for each k , the conditions of Theorem 5 hold, $(\hat{A}, \hat{B}, \hat{C}, \hat{D}) - (A, B, C, D) = o_p(1)$, $(\hat{\lambda}_k, \hat{\omega}_k) - (\lambda_k^*, \omega_k^*) = o_p(1)$,

$$\begin{aligned} \text{Var}_R(\Delta) &= O_p(1), & \text{Var}_R(\hat{g}) + \frac{n}{N} \text{Var}_O(r_0 \hat{g}) &= O_p(1), \\ \mathbb{P}(\hat{A} \leq t_A) + \mathbb{P}(\hat{B} \leq t_B) &\rightarrow 0, \end{aligned} \quad (45)$$

and the evaluation-step difference in item 2 of Proposition 7 is negligible at the $\sqrt{n}$ scale,

$$\hat{\omega}_k^2 \frac{n}{N} \text{Var}_O\{(\hat{r} - r_0) \hat{g}\} = o_p(1). \quad (46)$$

Then, for each k ,

$$\sqrt{n} \left\{ \hat{\theta}_{\hat{r},k} - \hat{\theta}_{r_0}(\lambda_k^*, \omega_k^*) \right\} = o_p(1), \quad (47)$$

so σ_k^2 equals the limit of $\tilde{V}_k$ in Equation (23) and

$$\sigma_{\text{joint}}^2 \leq \sigma_{\text{PPI}^{++}}^2 \leq \sigma_{\text{RCT}}^2. \quad (48)$$

This is not a universal finite-sample dominance statement.

4. Data Fusion for CATE Estimation

In this section, we develop two methods that combine an RCT with a potentially confounded OBS to estimate the CATE in the RCT population. The RCT identifies the causal target. The OBS outcome regressions and their treatment-effect prediction may improve statistical efficiency, but they need not be causally valid. The first method is the Doubly Robust Fusion Learner, based on an AIPW pseudo-outcome. The second is the R-Fusion Learner, based on an R-loss scaled by the known trial propensity variance.

4.1. CATE targets and RCT-valid learning losses

Definition 11 (CATE target, candidate class, and ideal risk). The target is $\tau(x) \triangleq \mathbb{E}_R\{Y(1) - Y(0) \mid X = x\}$. Let $\mathcal{Z} \subseteq L^2(P_R^X)$ be a prespecified nonempty class of candidate CATE functions, such as a fixed linear sieve, and let $\zeta \in \mathcal{Z}$. Its ideal risk is

$$\mathcal{R}(\zeta) \triangleq \mathbb{E}_R\{\zeta(X) - \tau(X)\}^2. \quad (49)$$

Throughout each conditional calculation, the nuisance fits are held fixed.

The first tuning coefficient blends the RCT and OBS outcome-regression pairs inside an RCT-valid score.

Definition 12 (Fused outcome nuisance and pseudo-outcome). For a fixed $\lambda \in [0, 1]$, define

$$\begin{aligned} q_R &\triangleq \hat{\mu}_R, \\ q_O &\triangleq \hat{\mu}_O, \\ q_\lambda &\triangleq (1 - \lambda)q_R + \lambda q_O, \\ Z_\lambda &\triangleq \varphi(V; q_\lambda). \end{aligned}$$

The AIPW score φ is defined in Equation (5).

Under Assumptions 2 and 1, Equation (6) gives, for every fixed λ ,

$$\mathbb{E}_R(Z_\lambda \mid X) = \tau(X).$$

Thus the known RCT propensity makes the fused pseudo-outcome conditionally valid without requiring either fitted outcome regression to be correct.

Proposition 8 (RCT DR loss and target). Assume RCT consistency and conditional randomization, a known trial propensity with strict overlap, and nuisance fits fixed independently of the loss-evaluation observations. For fixed $\lambda \in [0, 1]$, $Z_\lambda \in L^2(P_R)$, and $\zeta \in \mathcal{Z}$, define

$$\begin{aligned} \mathcal{L}_{\text{DR}}(\zeta; \lambda) &\triangleq \mathbb{E}_R\{Z_\lambda - \zeta(X)\}^2, \\ Q_{\text{DR}}(\lambda) &\triangleq \mathbb{E}_R\{Z_\lambda - \tau(X)\}^2. \end{aligned}$$

Then

$$\mathcal{L}_{\text{DR}}(\zeta; \lambda) = Q_{\text{DR}}(\lambda) + \mathcal{R}(\zeta).$$

Consequently,

$$\arg \min_{\zeta \in \mathcal{Z}} \mathcal{L}_{\text{DR}}(\zeta; \lambda) = \arg \min_{\zeta \in \mathcal{Z}} \mathcal{R}(\zeta).$$

Thus minimizing the population RCT DR loss over $\mathcal{Z}$ returns the best CATE approximation in $\mathcal{Z}$ under the ideal risk in Equation (49). The term $Q_{\text{DR}}(\lambda)$ is a ζ -independent noise floor, not the final risk of a fitted learner.

For the R-learner-style construction, define the known RCT propensity and the corresponding marginal-outcome nuisances using the same λ :

$$\begin{aligned} e_0(x) &\triangleq P_R(A = 1 \mid X = x), \\ m_R(x) &\triangleq e_0(x)q_{R,1}(x) + \{1 - e_0(x)\}q_{R,0}(x), \\ m_O(x) &\triangleq e_0(x)q_{O,1}(x) + \{1 - e_0(x)\}q_{O,0}(x), \\ m_\lambda(x) &\triangleq (1 - \lambda)m_R(x) + \lambda m_O(x). \end{aligned}$$

Proposition 9 (RCT R loss and target). Under Assumptions 2 and 1, let m be a fixed admissible measurable function, suppose $Y - m(X) \in L^2(P_R)$, and let $\zeta \in \mathcal{Z}$. Define

$$\begin{aligned} \mathcal{L}_R(\zeta; m) &\triangleq \mathbb{E}_R \left[\frac{[Y - m(X) - \{A - e_0(X)\}\zeta(X)]^2}{e_0(X)\{1 - e_0(X)\}} \right], \\ Q_R(m) &\triangleq \mathbb{E}_R \left[\frac{[Y - m(X) - \{A - e_0(X)\}\tau(X)]^2}{e_0(X)\{1 - e_0(X)\}} \right]. \end{aligned}$$

Suppose the displayed losses are integrable. Then

$$\mathcal{L}_R(\zeta; m) = Q_R(m) + \mathcal{R}(\zeta).$$

Consequently,

$$\arg \min_{\zeta \in \mathcal{Z}} \mathcal{L}_R(\zeta; m) = \arg \min_{\zeta \in \mathcal{Z}} \mathcal{R}(\zeta).$$

Thus minimizing the population RCT R loss over $\mathcal{Z}$ returns the same best CATE approximation under the ideal risk in Equation (49). In particular, $Q_R(m_\lambda)$ is the noise floor associated with the fused marginal-outcome nuisance. The two terms $Q_{\text{DR}}(\lambda)$ and $Q_R(m_\lambda)$ are different noise floors and need not select the same λ ; neither is the fitted CATE risk.

Both RCT-only routes identify the desired projection, but neither directly averages the large OBS covariate sample. Recall the fitted OBS treatment-effect prediction $\hat{g}$ from Equation (12), and define $G_\zeta(X) \triangleq \{\hat{g}(X) - \zeta(X)\}^2$. The following full-fusion objects are retained here so that the subsequent orthogonality and finite-sample results remain self-contained.

The primary *Doubly Robust Fusion (DRF) Learner* extends $\mathcal{L}_{\text{DR}}$ to use OBS covariates through the RCT-valid AIPW score. Here “doubly robust” names that score construction under known e_0 ; it does not by itself assert numerical-loss orthogonality or robustness to ratio error.

Definition 13 (Doubly Robust Fusion loss). For fixed ζ , define

$$\widehat{L}_{\text{DRF}}(\zeta; \lambda, \omega, r) \triangleq \mathbb{P}_R^{\text{fit}}\{Z_\lambda - \zeta(X)\}^2 + \omega \left[\mathbb{P}_O^{\text{fit}}\{r(X)G_\zeta(X)\} - \mathbb{P}_R^{\text{fit}}\{G_\zeta(X)\} \right]. \quad (50)$$

Assume $Z_\lambda \in L^2(P_R)$, $\zeta \in L^2(P_R^X)$, $G_\zeta \in L^1(P_R)$, and $r_0 G_\zeta \in L^1(P_O)$. The coefficient ω scales the OBS loss correction and is not part of G_ζ .

The first term in Equation (50) is the sample counterpart of $\mathcal{L}_{\text{DR}}(\zeta; \lambda)$. The coefficient ω multiplies the transported OBS loss correction, whose mean is zero when $r = r_0$.

Proposition 10 (Exact DRF target). Under Assumptions 2 and 1, suppose either Assumption 3 holds and $r_0 = 1$, or Assumption 4 holds and $r_0 = dP_R^X/dP_O^X$. Use $r = r_0$. For fixed λ, ω, ζ , suppose $Z_\lambda \in L_2(P_R)$, $\zeta \in L_2(P_R^X)$, $G_\zeta \in L_1(P_R)$, and $r_0 G_\zeta \in L_1(P_O)$. Then

$$\begin{aligned} \mathbb{E}\{\widehat{L}_{\text{DRF}}(\zeta; \lambda, \omega, r_0)\} &= \mathcal{L}_{\text{DR}}(\zeta; \lambda) \\ &= Q_{\text{DR}}(\lambda) + \mathcal{R}(\zeta). \end{aligned} \quad (51)$$

Thus, whenever the minimum is attained, the population minimizers over any fixed class $\mathcal{Z}$ are the $L_2(P_R^X)$ projections of τ . Under the common covariate marginal, $r_0 = 1$. The expected loss is flat in ω .

To leverage OBS covariates through the R route, the secondary *R-Fusion (RF) Learner* extends $\mathcal{L}_R(\zeta; m)$ with a transported loss correction.

Definition 14 (R-Fusion loss). For a fixed marginal-outcome function m , define

$$\kappa(A, X) \triangleq \frac{\{A - e_0(X)\}^2}{e_0(X)\{1 - e_0(X)\}}.$$

Define

$$\begin{aligned} \widehat{L}_{\text{RF}}(\zeta; m, \omega, r) &\triangleq \mathbb{P}_R^{\text{fit}} \left[\frac{[Y - m(X) - \{A - e_0(X)\}\zeta(X)]^2}{e_0(X)\{1 - e_0(X)\}} \right] \\ &\quad + \omega \left[\mathbb{P}_O^{\text{fit}}\{r(X)G_\zeta(X)\} - \mathbb{P}_R^{\text{fit}}\{\kappa(A, X)G_\zeta(X)\} \right]. \end{aligned} \quad (52)$$

Assume $\tau \in L^2(P_R)$ and the displayed RCT loss is integrable.

The first term in Equation (52) is the sample counterpart of $\mathcal{L}_R(\zeta; m)$. The factor $\kappa(A, X)$ is the RCT curvature weight induced by the scaled squared treatment residual, so the RCT subtraction matches the R-loss curvature. The correction has mean zero under exact transport.

Theorem 6 (Exact RF target). Assume RCT consistency and conditional randomization, a known trial propensity with strict overlap, and nuisance fits fixed independently of the fitting observations. Suppose either $P_R^X = P_O^X$ with $r_0 = 1$, or $P_R^X \ll P_O^X$ with exact $r_0 = dP_R^X/dP_O^X$. For fixed m, ζ, ω , suppose $\zeta, \tau \in L^2(P_R^X)$, the scaled RCT R-loss is in $L_1(P_R)$, $G_\zeta \in L_1(P_R)$, and $r_0 G_\zeta \in L_1(P_O)$. Suppose also that the minimum over the prespecified function class is attained whenever a minimizer is invoked. Then

$$\begin{aligned}\mathbb{E}\{\widehat{L}_{\text{RF}}(\zeta; m, \omega, r_0)\} &= \mathcal{L}_R(\zeta; m) \\ &= Q_R(m) + \mathcal{R}(\zeta).\end{aligned}\tag{53}$$

In particular, setting $m = m_\lambda$ gives the noise floor $Q_R(m_\lambda)$. Hence the population minimizers over a fixed class are the same unweighted $L_2(P_R^X)$ projections of τ .

4.2. Orthogonality of learning equations and nuisance sensitivity

Proposition 10 and Theorem 6 establish the exact population targets. We now ask a different question: how does each fitted object enter the first-order equation that learns ζ ?

Definition 15 (Derivative directions and learning roles). For a functional F of a function-valued argument η , let h be an admissible direction such that $\eta + \varepsilon h$ remains in the domain of F for all sufficiently small $|\varepsilon|$. Whenever the derivative exists, define the Gateaux derivative by

$$D_\eta F(\eta)[h] \triangleq \left. \frac{d}{d\varepsilon} F(\eta + \varepsilon h) \right|_{\varepsilon=0}.\tag{54}$$

The target argument is ζ , and v denotes an admissible target direction. The learned objects $q, m, \widehat{g}, r$ are nuisance arguments. The coefficients λ and ω are held fixed during this audit: λ selects q_λ or m_λ , while ω scales the OBS correction. The trial propensity e_0 is known by design.

Following Foster and Syrgkanis (2023, Assumption 1), for a population loss $L(\zeta; \eta)$ with target argument ζ and nuisance argument η , loss orthogonality means that the mixed derivative $D_\eta D_\zeta L$ vanishes at the population target and nuisance truth. Equivalently, up to scaling, it means $D_\eta \Psi = 0$ for $\Psi \triangleq -D_\zeta L/2$. This definition alone does not establish an oracle convergence rate.

Write $Z_q \triangleq \varphi(V; q)$, $s_0(X) \triangleq e_0(X)\{1 - e_0(X)\}$, and $\kappa_0(A, X) \triangleq \kappa(A, X) = \{A - e_0(X)\}^2/s_0(X)$.

Definition 16 (Full fusion learning equations). For fixed (λ, ω) , define the DRF and RF population learning equations by

$$\begin{aligned}\Psi_{\text{DRF}}(\zeta; q, \widehat{g}, r)[v] &\triangleq -\frac{1}{2} D_\zeta \mathbb{E}\{\widehat{L}_{\text{DRF}}\}[v] \\ &= \mathbb{E}_R[v(X)\{Z_q - \zeta(X)\}] \\ &\quad + \omega [\mathbb{E}_O\{r(X)v(X)[\widehat{g}(X) - \zeta(X)]\} - \mathbb{E}_R\{v(X)[\widehat{g}(X) - \zeta(X)]\}],\end{aligned}\tag{55}$$

$$\begin{aligned}
\Psi_{\text{RF}}(\zeta; m, \hat{g}, r)[v] &\triangleq -\frac{1}{2}D_\zeta \mathbb{E}\{\hat{L}_{\text{RF}}\}[v] \\
&= \mathbb{E}_R \left[v(X) \frac{A - e_0(X)}{s_0(X)} \{Y - m(X) - [A - e_0(X)]\zeta(X)\} \right] \\
&\quad + \omega [\mathbb{E}_O\{r(X)v(X)[\hat{g}(X) - \zeta(X)]\} - \mathbb{E}_R\{\kappa_0(A, X)v(X)[\hat{g}(X) - \zeta(X)]\}].
\end{aligned} \tag{56}$$

All derivatives hold the remaining displayed arguments fixed.

Proposition 11 (Exact-transport learning equations). Assume RCT consistency and conditional randomization, a known propensity e_0 satisfying strict overlap, and nuisance functions fitted honestly on data independent of the loss-evaluation observations. Suppose either that the RCT and OBS have a common covariate marginal, in which case $r_0 = 1$, or that exact covariate transport holds with $r_0 = dP_R^X/dP_O^X$. Use $r = r_0$, and suppose all displayed expectations below are finite. For every integrable function $f(X)$, exact transport gives $\mathbb{E}_O(r_0 f) = \mathbb{E}_R(f)$, and randomization gives $\mathbb{E}_R(\kappa_0 | X) = 1$. Consequently,

$$\begin{aligned}
\Psi_{\text{DRF}}(\zeta; q, \hat{g}, r_0)[v] &= \mathbb{E}_R[v(X)\{Z_q - \zeta(X)\}], \\
\Psi_{\text{RF}}(\zeta; m, \hat{g}, r_0)[v] &= \mathbb{E}_R \left[v(X) \frac{A - e_0(X)}{s_0(X)} \{Y - m(X) - [A - e_0(X)]\zeta(X)\} \right].
\end{aligned}$$

Thus exact transport removes the OBS correction from the population learning equation without removing it from the empirical optimization problem.

Proposition 12 (Direction-by-direction learning-equation orthogonality). Assume honest fitting, strict trial overlap, the integrability needed to interchange differentiation and expectation, and exact fixed r_0 where stated. Then the following partial-path statements hold.

1. For an outcome-regression direction $h = (h_0, h_1)$,

$$\begin{aligned}
\dot{Z}_q[h] &\triangleq -\{A - e_0(X)\} \left\{ \frac{h_1(X)}{e_0(X)} + \frac{h_0(X)}{1 - e_0(X)} \right\}, \\
D_q \Psi_{\text{DRF}}[h, v] &= \mathbb{E}_R\{v(X)\dot{Z}_q[h]\} = 0.
\end{aligned}$$

This exact score orthogonality is the main protection supplied by the known trial propensity.

2. For a marginal-outcome direction h ,

$$D_m \Psi_{\text{RF}}[h, v] = -\mathbb{E}_R \left[v(X) \frac{A - e_0(X)}{s_0(X)} h(X) \right] = 0.$$

3. For a prediction direction u , define the path $\hat{g}_\varepsilon \triangleq \hat{g} + \varepsilon u$. For either $j \in \{\text{DRF}, \text{RF}\}$, write $\Psi_j(\hat{g}_\varepsilon)[v]$ and $\hat{L}_j(\hat{g}_\varepsilon)$ with all other displayed arguments held fixed. By Equation (54), the two derivatives below are $D_{\hat{g}} \Psi_j[u, v]$ and $D_{\hat{g}} \mathbb{E}\{\hat{L}_j\}[u]$, in the same nota-

tion as items 1 to 3 and 5. Then

$$\left. \frac{d}{d\varepsilon} \Psi_j(\widehat{g}_\varepsilon)[v] \right|_{\varepsilon=0} = \omega \mathbb{E}_O[\{r(X) - r_0(X)\}u(X)v(X)], \quad (57)$$

$$\left. \frac{d}{d\varepsilon} \mathbb{E}\{\widehat{L}_j(\widehat{g}_\varepsilon)\} \right|_{\varepsilon=0} = 2\omega \mathbb{E}_O[\{r(X) - r_0(X)\}\{\widehat{g}(X) - \zeta(X)\}u(X)]. \quad (58)$$

Both derivatives vanish when $r = r_0$, but not generally for an approximate ratio.

4. For a ratio direction h_r ,

$$D_r \Psi_j[h_r, v] = \omega \mathbb{E}_O[h_r(X)v(X)\{\widehat{g}(X) - \zeta(X)\}], \quad (59)$$

$$D_r \mathbb{E}\{\widehat{L}_j\}[h_r] = \omega \mathbb{E}_O[h_r(X)\{\widehat{g}(X) - \zeta(X)\}^2]. \quad (60)$$

These derivatives are generally nonzero.

The normalization constraint on a density ratio restricts its regular tangent directions. Define

$$\mathcal{T}_r \triangleq L_0^2(P_O^X) = \{h \in L^2(P_O^X) : \mathbb{E}_O(h) = 0\}.$$

This linear space describes first-order normalization; admissible paths must also respect support and nonnegativity, and a ratio on the positivity boundary may have only a tangent cone.

Proposition 13 (Density-ratio nonorthogonality). Suppose $\omega \neq 0$ and $v(\widehat{g} - \zeta) \in L^2(P_O^X)$. Over the full normalized tangent space $\mathcal{T}_r$, the ratio-score derivative in Equation (59) vanishes for every $h_r \in \mathcal{T}_r$ if and only if $v(X)\{\widehat{g}(X) - \zeta(X)\}$ is constant P_O^X -almost surely. Therefore, the learning equations are generally not orthogonal to the density ratio. Balancing selected empirical moments does not generally produce orthogonality over the full normalized tangent space.

As a direct chain-rule consequence of items 1–3 in Proposition 12, if the same OBS outcome regression generates $(q_O, \widehat{g})$ for DRF or $(m_O, \widehat{g})$ for RF, then its total learning-equation derivative is zero when e_0 is known and exact r_0 is held fixed. This conclusion does not cover paths that change r , or the use of an approximate ratio.

These results establish the target and its first-order nuisance sensitivity, but they do not control the error of the fitted CATE function. We first derive that error for a general function class with explicit RCT and OBS fitting sample sizes. We then select (λ, ω) by honest validation.

4.3. General two-source learning theory

Fix a method $j \in \{\text{DRF}, \text{RF}\}$, coefficients (λ, ω) , and honestly fitted nuisances. Let n_f and N_f denote the independent RCT and OBS fitting-sample sizes. The method-specific losses in Equations (50) and (52) can be written as an RCT contribution $\ell_{R,j}(\zeta)$ and an OBS contribution $\ell_{O,j}(\zeta)$.

Definition 17 (General fusion empirical learner). For $\zeta \in \mathcal{Z}$, define

$$\begin{aligned} L_j(\zeta) &\triangleq P_R \ell_{R,j}(\zeta) + P_O \ell_{O,j}(\zeta), \\ \widehat{L}_j(\zeta) &\triangleq \mathbb{P}_{R,n_f} \ell_{R,j}(\zeta) + \mathbb{P}_{O,N_f} \ell_{O,j}(\zeta). \end{aligned}$$

Let

$$\zeta_j^\circ \in \arg \min_{\zeta \in \mathcal{Z}} \mathcal{R}(\zeta)$$

be a prespecified, deterministically selected best-in-class function. Let $\text{Pen} : \mathcal{Z} \rightarrow [0, \infty]$ be the full penalty functional and define

$$\widehat{J}_j(\zeta) \triangleq \widehat{L}_j(\zeta) + \text{Pen}(\zeta).$$

Assume $\widehat{J}_j$ is bounded below and has a measurable global ϵ_{opt} -approximate minimizer $\widehat{\zeta}_j$, meaning

$$\widehat{J}_j(\widehat{\zeta}_j) \leq \inf_{\zeta \in \mathcal{Z}} \widehat{J}_j(\zeta) + \epsilon_{\text{opt}}.$$

Finally, define the centered comparator process

$$\mathbb{Z}_j(\zeta) \triangleq \{\widehat{L}_j - L_j\}(\zeta) - \{\widehat{L}_j - L_j\}(\zeta_j^\circ).$$

The next theorem is deterministic once the two fitting samples and fitted nuisances are fixed. It is the master reduction from fusion-loss fluctuation to CATE excess risk.

Theorem 7 (Exact general fusion ERM reduction). Assume RCT consistency and conditional randomization, a known propensity e_0 satisfying strict overlap, and honest nuisance fitting. Suppose either $P_R^X = P_O^X$ and $r_0 = 1$, or $P_R^X \ll P_O^X$ and the exact ratio $r_0 = dP_R^X/dP_O^X$ is used. Suppose the displayed losses are finite and $\widehat{J}_j$ is bounded below and has a measurable global ϵ_{opt} -approximate minimizer $\widehat{\zeta}_j$, meaning $\widehat{J}_j(\widehat{\zeta}_j) \leq \inf_{\zeta \in \mathcal{Z}} \widehat{J}_j(\zeta) + \epsilon_{\text{opt}}$. Then

$$\mathcal{R}(\widehat{\zeta}_j) - \mathcal{R}(\zeta_j^\circ) \leq -\mathbb{Z}_j(\widehat{\zeta}_j) + \text{Pen}(\zeta_j^\circ) - \text{Pen}(\widehat{\zeta}_j) + \epsilon_{\text{opt}}. \quad (61)$$

Consequently, the right-hand side is at most $\sup_{\zeta \in \mathcal{Z}} \{-\mathbb{Z}_j(\zeta)\} + \text{Pen}(\zeta_j^\circ) + \epsilon_{\text{opt}}$.

To obtain a sample-size-dependent result, localize by excess risk rather than by an unnamed norm. For $u \geq 0$, define

$$\mathcal{Z}_j(u) \triangleq \{\zeta \in \mathcal{Z} : \mathcal{R}(\zeta) - \mathcal{R}(\zeta_j^\circ) \leq u\}.$$

Let $\mathbb{Z}_{R,j}$ and $\mathbb{Z}_{O,j}$ be the sourcewise centered increments whose sum is $\mathbb{Z}_j$. Suppose measurable functions $\psi_{R,j}, \psi_{O,j}$ and $\text{Tail}_{R,j}, \text{Tail}_{O,j}$ satisfy, sourcewise and uniformly over the localized class, the high-probability upper bounds stated in the theorem below. Define

$$\begin{aligned} \Phi_{j,n_f,N_f}(u; \delta) &\triangleq \psi_{R,j}(u; n_f) + \psi_{O,j}(u; N_f) + \text{Tail}_{R,j}(u, \delta) + \text{Tail}_{O,j}(u, \delta), \\ u_{j,n_f,N_f}^*(\delta) &\triangleq \inf\{u > 0 : \Phi_{j,n_f,N_f}(u; \delta) \leq c_0 u\}, \quad 0 < c_0 < 1. \end{aligned}$$

For every fixed δ , assume that the full function $u \mapsto \Phi_{j,n_f,N_f}(u; \delta)$ is nondecreasing and that $u \mapsto \Phi_{j,n_f,N_f}(u; \delta)/\sqrt{u}$ is nonincreasing on $(0, \infty)$. Thus the full RCT-plus-OBS modulus, including its tail terms, has the sub-root scaling used below. Thus u_{j,n_f,N_f}^* is an excess-risk level, not an L_2 -distance radius.

Theorem 8 (Localized two-source excess risk). Assume RCT consistency and conditional randomization, a known strict-overlap propensity e_0 , honest fixed nuisances, and either a common covariate marginal with $r_0 = 1$ or exact transport with $r_0 = dP_R^X/dP_O^X$. Assume the displayed losses are finite, the penalized objective is bounded below, and a measurable global ϵ_{opt} -approximate minimizer exists. Assume the RCT and OBS fitting samples are independent; the function class and localized loss-increment classes are measurable or separable; their envelopes are finite. For every fixed δ , assume the full modulus $u \mapsto \Phi_{j,n_f,N_f}(u; \delta)$ is nondecreasing and $u \mapsto \Phi_{j,n_f,N_f}(u; \delta)/\sqrt{u}$ is nonincreasing for $u > 0$. Assume also that, with probability at least $1 - \delta$ for each source,

$$\begin{aligned} \sup_{\zeta \in \mathcal{Z}_j(u)} \{-\mathbb{Z}_{R,j}(\zeta)\} &\leq \psi_{R,j}(u; n_f) + \text{Tail}_{R,j}(u, \delta), \\ \sup_{\zeta \in \mathcal{Z}_j(u)} \{-\mathbb{Z}_{O,j}(\zeta)\} &\leq \psi_{O,j}(u; N_f) + \text{Tail}_{O,j}(u, \delta) \end{aligned}$$

simultaneously on the peeling levels used in the proof. Then, with probability at least $1 - 2\delta$,

$$\mathcal{R}(\hat{\zeta}_j) - \mathcal{R}(\zeta_j^\circ) \leq C_0 \left\{ u_{j,n_f,N_f}^*(\delta) + \text{Pen}(\zeta_j^\circ) + \epsilon_{\text{opt}} \right\}, \quad (62)$$

where C_0 depends only on the stated sub-root and fixed-point constants. The theorem does not assert empirical convexity; the global approximate-ERM condition remains substantive when the source correction is signed.

An estimated density ratio changes both the population loss and the empirical process. Condition on the ratio-training sample and let $u_{j,n_f,N_f}^*(\hat{r}; \delta)$ denote the preceding fixed point for the $\hat{r}$ -weighted empirical loss. Define the fitting drift

$$\Delta_j^{\text{fit}}(u; \hat{r}) \triangleq \sup_{\zeta \in \mathcal{Z}_j(u)} \left| \{L_j^{\hat{r}} - L_j^{r_0}\}(\zeta) - \{L_j^{\hat{r}} - L_j^{r_0}\}(\zeta_j^\circ) \right|.$$

Proposition 14 (General estimated-ratio fitting drift). Assume RCT consistency and conditional randomization, a known strict-overlap propensity, honest nuisance fitting, independent RCT and OBS fitting samples, and an independent ratio-training sample. Suppose either $P_R^X = P_O^X$ with $r_0 = 1$, or $P_R^X \ll P_O^X$ with $r_0 = dP_R^X/dP_O^X$. Condition on the ratio-training sample. Assume the actual $\hat{r}$ -weighted penalized objective is bounded below and has a measurable global approximate minimizer, and that its measurable localized classes have finite envelopes and sourcewise high-probability modulus bounds. For every fixed δ , assume their full RCT-plus-OBS modulus $\Phi(u; \delta)$ is nondecreasing and $\Phi(u; \delta)/\sqrt{u}$ is nonincreasing for $u > 0$, including the stated tail terms. Form the fixed point from the actual $\hat{r}$ -weighted loss. Assume the localization closes at $\bar{u}$ in the explicit sense that

$$C_1 \{ u_{j,n_f,N_f}^*(\hat{r}; \delta) + \Delta_j^{\text{fit}}(\bar{u}; \hat{r}) + \text{Pen}(\zeta_j^\circ) + \epsilon_{\text{opt}} \} \leq \bar{u}.$$

Then, with probability at least $1 - 2\delta$ conditional on the ratio-training sample,

$$\mathcal{R}(\widehat{\zeta}_j^r) - \mathcal{R}(\zeta_j^\circ) \leq C_1 \left\{ u_{j,n_f,N_f}^*(\widehat{r}; \delta) + \Delta_j^{\text{fit}}(\bar{u}; \widehat{r}) + \text{Pen}(\zeta_j^\circ) + \epsilon_{\text{opt}} \right\}.$$

If $\|\widehat{r} - r_0\|_{L^2(P_O)} \leq a_{n_r, N_r}$ and

$$\sup_{\zeta \in \mathcal{Z}_j(u)} \|G_\zeta - G_{\zeta_j^\circ}\|_{L^2(P_O)} \leq B_{G,j}(u),$$

then $\Delta_j^{\text{fit}}(u; \widehat{r}) \leq |\omega| a_{n_r, N_r} B_{G,j}(u)$. This is a fitting-drift result, not density-ratio orthogonality.

The preceding results control each fixed candidate. We now select the fusion coefficients while keeping the final validation observations separate.

4.4. Choosing the fusion coefficients

Definition 18 (Fitted fusion learner and final-risk oracle). Let $j \in \{\text{DRF}, \text{RF}\}$ label the Doubly Robust Fusion Learner or the R-Fusion Learner, and let $\Omega \subset \mathbb{R}$ be a prespecified compact set containing zero. For fixed $(\lambda, \omega) \in [0, 1] \times \Omega$, define $\widehat{\tau}_j(\cdot; \lambda, \omega) \in \mathcal{Z}$ as the output of a prespecified empirical minimization rule, with deterministic tie-breaking, applied to the empirical DRF loss in Equation (50) or the empirical RF loss in Equation (52), respectively, on the fitting sample. After the nuisance fits, fitting sample, and fitting-stage algorithmic randomness are fixed, define its final fitted risk over a fresh $X \sim P_R^X$ by

$$\mathcal{R}\{\widehat{\tau}_j(\cdot; \lambda, \omega)\} \triangleq \mathbb{E}_R \left[\{\widehat{\tau}_j(X; \lambda, \omega) - \tau(X)\}^2 \mid \text{fitted objects} \right].$$

Whenever the minimum is attained, define the infeasible final-risk oracle by

$$(\lambda_j^*, \omega_j^*) \in \arg \min_{\lambda \in [0, 1], \omega \in \Omega} \mathcal{R}\{\widehat{\tau}_j(\cdot; \lambda, \omega)\}. \quad (63)$$

In DRF, λ changes Z_λ ; in RF, it changes m_λ . In both methods, ω changes the empirical transported OBS correction. Hence the fitted function depends on both coefficients, even though exact transport cancels the correction in conditional expectation for each fixed candidate function. Recall from Equation (49) that for any fixed function f , $\mathcal{R}(f) = \mathbb{E}_R[\{f(X) - \tau(X)\}^2]$. Throughout this section, an efficiency gain means a reduction in the integrated squared CATE error $\mathcal{R}$ of the fitted learner. This operational learning criterion is distinct from attaining a semiparametric efficiency bound. OBS information may reduce this error without changing the RCT-population target. This oracle is infeasible because τ is unknown, and its value changes across fitting-sample realizations. For both methods, $(\lambda, \omega) = (0, 0)$ gives the corresponding RCT-only learner. Since this pair is feasible, the oracle obeys the conditional comparison

$$\mathcal{R}\{\widehat{\tau}_j(\cdot; \lambda_j^*, \omega_j^*)\} \leq \mathcal{R}\{\widehat{\tau}_j(\cdot; 0, 0)\}. \quad (64)$$

This is an oracle comparison, not a guarantee for estimated coefficients. There is no universal closed form for Equation (63).

Two simpler closed forms minimize the base-loss noise floors rather than the fitted risk. Conditional on honestly fitted and now fixed nuisances, the fixed-function decompositions from Propositions 8 and 9 are

$$\mathcal{L}_{\text{DR}}(\zeta; \lambda) = Q_{\text{DR}}(\lambda) + \mathcal{R}(\zeta), \quad \mathcal{L}_R(\zeta; m_\lambda) = Q_R(m_\lambda) + \mathcal{R}(\zeta).$$

Recall that $Z_0 = \varphi(V; q_R)$ and $Z_1 = \varphi(V; q_O)$, while m_R and m_O are the corresponding RCT and OBS marginal-outcome nuisances. Propositions 8 and 9 define $Q_{\text{DR}}(\lambda)$ and $Q_R(m_\lambda)$, respectively. These terms do not equal the ideal risk of a fitted CATE function.

Proposition 15 (Closed-form noise-floor coefficients). Assume RCT consistency and conditional randomization, a known propensity e_0 satisfying strict overlap, and nuisance functions fitted honestly on data independent of the observations used below. Condition on these fixed nuisance fits and suppose the displayed expectations are finite. Let $q_R = (q_{R,0}, q_{R,1})$ and $q_O = (q_{O,0}, q_{O,1})$ be the fixed RCT and OBS outcome-regression pairs, and define

$$\begin{aligned} Z_0 &\triangleq \varphi(V; q_R), & Z_1 &\triangleq \varphi(V; q_O), \\ m_R(X) &\triangleq e_0(X)q_{R,1}(X) + \{1 - e_0(X)\}q_{R,0}(X), \\ m_O(X) &\triangleq e_0(X)q_{O,1}(X) + \{1 - e_0(X)\}q_{O,0}(X), \\ \Delta &\triangleq Z_1 - Z_0, & d_m(X) &\triangleq m_O(X) - m_R(X). \end{aligned}$$

For $Z_\lambda \triangleq Z_0 + \lambda\Delta$ and $m_\lambda \triangleq m_R + \lambda d_m$, define

$$\begin{aligned} Q_{\text{DR}}(\lambda) &\triangleq \mathbb{E}_R\{Z_\lambda - \tau(X)\}^2, \\ Q_R(m_\lambda) &\triangleq \mathbb{E}_R \left[\frac{[Y - m_\lambda(X) - \{A - e_0(X)\}\tau(X)]^2}{e_0(X)\{1 - e_0(X)\}} \right]. \end{aligned}$$

For an interval, the projection used below is $\text{Proj}_{[0,1]}(u) \triangleq \min\{1, \max(0, u)\}$. If $\mathbb{E}_R(\Delta^2) > 0$, the constrained DR noise-floor minimizer is

$$\lambda_{\text{DR},Q}^* \triangleq \text{Proj}_{[0,1]} \left\{ -\frac{\mathbb{E}_R(Z_0\Delta)}{\mathbb{E}_R(\Delta^2)} \right\}. \quad (65)$$

If $\mathbb{E}_R[d_m(X)^2/\{e_0(X)(1 - e_0(X))\}] > 0$, the constrained R noise-floor minimizer is

$$\lambda_{R,Q}^* \triangleq \text{Proj}_{[0,1]} \left[\frac{\mathbb{E}_R[\{Y - m_R(X)\}d_m(X)/\{e_0(X)(1 - e_0(X))\}]}{\mathbb{E}_R[d_m(X)^2/\{e_0(X)(1 - e_0(X))\}]} \right]. \quad (66)$$

If either denominator is zero, its corresponding noise floor is flat in λ , so every $\lambda \in [0, 1]$ is a minimizer. The displayed equivalent moments do not involve τ .

The two coefficients in Proposition 15 are population noise-floor oracles. They are neither fitted-risk optimizers in Equation (63) nor data-adaptive coefficient estimates.

Proposition 16 (Population-loss flatness in the fusion scale). Assume RCT consistency and conditional randomization, a known trial propensity e_0 satisfying strict overlap, and nuisance functions fitted honestly on data independent of the loss-evaluation observations. Suppose

either that the RCT and OBS have a common covariate marginal with $r_0 = 1$, or that exact covariate transport holds with $r_0 = dP_R^X/dP_O^X$. Use $r = r_0$, let $\Omega \subset \mathbb{R}$ be nonempty and compact, and fix $\zeta \in \mathcal{Z}$ and $\lambda \in [0, 1]$. Condition on the fitted nuisances and suppose the expectations below are finite. For every $\omega \in \Omega$, the method-specific corrections satisfy

$$\begin{aligned}\mathbb{E}_O\{r_0(X)G_\zeta(X)\} - \mathbb{E}_R\{G_\zeta(X)\} &= 0, \\ \mathbb{E}_O\{r_0(X)G_\zeta(X)\} - \mathbb{E}_R\{\kappa(A, X)G_\zeta(X)\} &= 0.\end{aligned}$$

Consequently,

$$\begin{aligned}\mathbb{E}\{\widehat{L}_{\text{DRF}}(\zeta; \lambda, \omega, r_0)\} &= \mathcal{L}_{\text{DR}}(\zeta; \lambda), \\ \mathbb{E}\{\widehat{L}_{\text{RF}}(\zeta; m_\lambda, \omega, r_0)\} &= \mathcal{L}_R(\zeta; m_\lambda),\end{aligned}$$

and both right-hand sides are constant in ω . Therefore,

$$\begin{aligned}\arg \min_{\omega \in \Omega} \mathbb{E}\{\widehat{L}_{\text{DRF}}(\zeta; \lambda, \omega, r_0)\} &= \Omega, \\ \arg \min_{\omega \in \Omega} \mathbb{E}\{\widehat{L}_{\text{RF}}(\zeta; m_\lambda, \omega, r_0)\} &= \Omega.\end{aligned}$$

Thus population loss at a fixed function and fixed λ cannot select a unique fusion scale.

Although the conditional expected full-fusion loss at fixed ζ and fixed λ is flat in ω under exact r_0 , the fitted function depends on ω ; we therefore compare fully fitted coefficient-grid functions by honest validation, after all fitted objects are fixed.

Definition 19 (Honest finite-grid candidates). Fix $j \in \{\text{DRF}, \text{RF}\}$, its function class, and its penalty. Let $\mathcal{G}_j \subseteq [0, 1] \times \Omega$ be a prespecified nonempty finite grid containing $(0, 0)$, and fit $\widehat{\tau}_j(\cdot; \lambda, \omega)$ before validation for every $(\lambda, \omega) \in \mathcal{G}_j$. Define the infeasible grid oracle by

$$(\lambda_j^{\mathcal{G}}, \omega_j^{\mathcal{G}}) \in \arg \min_{(\lambda, \omega) \in \mathcal{G}_j} \mathcal{R}\{\widehat{\tau}_j(\cdot; \lambda, \omega)\}.$$

The continuous oracle in Equation (63) searches all of $[0, 1] \times \Omega$; this grid oracle searches only $\mathcal{G}_j$; the empirical selector below minimizes an observed validation criterion.

Definition 20 (Common-reference validation score). Set $Z_0 \triangleq \varphi(V; q_R)$, where q_R is fixed before validation, and let $\widehat{g}$ be the fitted OBS treatment-effect prediction in Equation (12). Under a common covariate marginal use $r_0 = 1$; under covariate shift use the exact ratio $r_0 = dP_R^X/dP_O^X$. Let $\mathbb{P}_{R, \text{val}}$ and $\mathbb{P}_{O, \text{val}}$ denote empirical averages over the RCT and OBS validation samples. Define

$$\begin{aligned}G_{j, \lambda, \omega}(X) &\triangleq \{\widehat{g}(X) - \widehat{\tau}_j(X; \lambda, \omega)\}^2, \\ S_{R, j, \lambda, \omega} &\triangleq \{Z_0 - \widehat{\tau}_j(X; \lambda, \omega)\}^2 - \omega G_{j, \lambda, \omega}(X), \\ S_{O, j, \lambda, \omega} &\triangleq \omega r_0(X) G_{j, \lambda, \omega}(X).\end{aligned}$$

The validation criterion is

$$\widehat{C}_j(\lambda, \omega) \triangleq \mathbb{P}_{R, \text{val}} S_{R,j,\lambda,\omega} + \mathbb{P}_{O, \text{val}} S_{O,j,\lambda,\omega}. \quad (67)$$

Equivalently, this sample quantity is the RCT pseudo-outcome loss plus $\omega\{\mathbb{P}_{O, \text{val}}(r_0 G_{j,\lambda,\omega}) - \mathbb{P}_{R, \text{val}}(G_{j,\lambda,\omega})\}$. The minus sign creates a transported OBS-minus-RCT control variate: under exact transport its conditional expected correction is zero. A plus sign would not have this property. Using deterministic tie-breaking, define

$$(\widehat{\lambda}_j, \widehat{\omega}_j) \in \arg \min_{(\lambda, \omega) \in \mathcal{G}_j} \widehat{C}_j(\lambda, \omega).$$

Proposition 17 (Common-noise validation identity). Assume RCT consistency and conditional randomization, a known trial propensity e_0 satisfying strict overlap, and honest nuisance fitting. Condition on the fitted grid functions and fitting randomness, and suppose the validation observations are independent of these objects. Suppose either $P_R^X = P_O^X$ and $r_0 = 1$, or $P_R^X \ll P_O^X$ and $r_0 = dP_R^X/dP_O^X$. Fix q_R before validation, so Equation (6) gives $\mathbb{E}_R(Z_0 | X) = \tau(X)$, and suppose the displayed expectations are finite. Then, for every $(\lambda, \omega) \in \mathcal{G}_j$,

$$\mathbb{E}_{\text{val}}[\widehat{C}_j(\lambda, \omega) | \text{fitted objects}] = Q_{\text{DR}}(0) + \mathcal{R}\{\widehat{\tau}_j(\cdot; \lambda, \omega)\}, \quad (68)$$

where $Q_{\text{DR}}(0) \triangleq \mathbb{E}_R\{Z_0 - \tau(X)\}^2$. This is a conditional-expectation identity over fresh validation observations, not an equality for each realized validation sample and not an unbiased estimate of $\mathcal{R}$ itself. It identifies candidate risk differences because the additive term $Q_{\text{DR}}(0)$ is common to every candidate.

Algorithm 2: Honest selection of CATE fusion coefficients

Input: The objects in Definition 20 and independent nuisance, fitting, and validation samples.

- 1 Fit the nuisances on the nuisance sample.
- 2 For every $(\lambda, \omega) \in \mathcal{G}_j$, fit $\widehat{\tau}_j(\cdot; \lambda, \omega)$ on the fitting sample.
- 3 Compute $\widehat{C}_j(\lambda, \omega)$ in Equation (67) for every grid pair on the common validation samples.
- 4 Select $(\widehat{\lambda}_j, \widehat{\omega}_j)$ by minimizing $\widehat{C}_j$ over $\mathcal{G}_j$.

Output: $(\widehat{\lambda}_j, \widehat{\omega}_j)$ and $\widehat{\tau}_j(\cdot; \widehat{\lambda}_j, \widehat{\omega}_j)$.

Let $M \triangleq |\mathcal{G}_j|$, let n_v, N_v be the RCT and OBS validation sizes, and suppose the RCT and OBS source-score ranges have lengths at most B_R and B_O . Define

$$\varepsilon_{\text{val}}(\delta) \triangleq B_R \sqrt{\frac{\log(4M/\delta)}{2n_v}} + B_O \sqrt{\frac{\log(4M/\delta)}{2N_v}}.$$

This is a uniform finite-grid validation-error radius obtained from Hoeffding concentration and a union bound over the M candidates. It is not a Brownian or Gaussian-process radius.

Theorem 9 (Finite-grid honest validation). Use the objects in Definition 20. Assume RCT consistency and conditional randomization, a known propensity e_0 satisfying strict overlap, and honest nuisance fitting. Condition on the finite grid, all fitted functions, and fitting randomness, all of which must be independent of final validation. Let the RCT and OBS validation samples be mutually independent. Suppose either $P_R^X = P_O^X$ and $r_0 = 1$, or $P_R^X \ll P_O^X$ and $r_0 = dP_R^X/dP_O^X$. Fix q_R before validation so that $\mathbb{E}_R(Z_0 | X) = \tau(X)$. Suppose the source scores $S_{R,j,\lambda,\omega}$ and $S_{O,j,\lambda,\omega}$ have uniform range lengths at most B_R and B_O , respectively, over $\mathcal{G}_j$. Then, with probability at least $1 - \delta$,

$$\mathcal{R}\{\hat{\tau}_j(\cdot; \hat{\lambda}_j, \hat{\omega}_j)\} \leq \mathcal{R}\{\hat{\tau}_j(\cdot; \lambda_j^{\mathcal{G}}, \omega_j^{\mathcal{G}})\} + 2\varepsilon_{\text{val}}(\delta) \leq \mathcal{R}\{\hat{\tau}_j(\cdot; 0, 0)\} + 2\varepsilon_{\text{val}}(\delta). \quad (69)$$

This is not realized-sample universal dominance. It also does not establish coefficient consistency, approximation to the continuous oracle, strict improvement, or unconditional dominance.

Corollary 2 (Estimated-ratio validation). Use the objects in Definition 20. Assume RCT consistency and conditional randomization, a known propensity e_0 satisfying strict overlap, and honest nuisance fitting. Condition on the finite grid, all fitted functions, and fitting randomness, all independent of mutually independent RCT and OBS final-validation samples. Suppose either $P_R^X = P_O^X$ and $r_0 = 1$, or $P_R^X \ll P_O^X$ and $r_0 = dP_R^X/dP_O^X$. Fix q_R before validation so that $\mathbb{E}_R(Z_0 | X) = \tau(X)$. Let an evaluation-independent $\hat{r}$ replace r_0 in the OBS validation score, and condition on $\hat{r}$. Define

$$\begin{aligned} \hat{S}_{O,j,\lambda,\omega}^{\hat{r}} &\triangleq \omega \hat{r}(X) G_{j,\lambda,\omega}(X), \\ \hat{C}_j^{\hat{r}}(\lambda, \omega) &\triangleq \mathbb{P}_{R,\text{val}} S_{R,j,\lambda,\omega} + \mathbb{P}_{O,\text{val}} \hat{S}_{O,j,\lambda,\omega}^{\hat{r}}. \end{aligned}$$

Suppose the actual RCT and OBS scores have uniform range lengths at most $\hat{B}_R$ and $\hat{B}_O$, and define

$$\begin{aligned} \hat{\varepsilon}_{\text{val}}(\delta) &\triangleq \hat{B}_R \sqrt{\frac{\log(4M/\delta)}{2n_v}} + \hat{B}_O \sqrt{\frac{\log(4M/\delta)}{2N_v}}, \\ \Delta_r &\triangleq \max_{(\lambda,\omega) \in \mathcal{G}_j} |\omega \mathbb{E}_O[(\hat{r} - r_0) G_{j,\lambda,\omega}]|. \end{aligned} \quad (70)$$

Let $(\hat{\lambda}_j^{\hat{r}}, \hat{\omega}_j^{\hat{r}})$ be the deterministically tie-broken minimizer of $\hat{C}_j^{\hat{r}}$ over $\mathcal{G}_j$. Then, with probability at least $1 - \delta$,

$$\begin{aligned} \mathcal{R}\{\hat{\tau}_j(\cdot; \hat{\lambda}_j^{\hat{r}}, \hat{\omega}_j^{\hat{r}})\} &\leq \mathcal{R}\{\hat{\tau}_j(\cdot; \lambda_j^{\mathcal{G}}, \omega_j^{\mathcal{G}})\} + 2\hat{\varepsilon}_{\text{val}}(\delta) + 2\Delta_r \\ &\leq \mathcal{R}\{\hat{\tau}_j(\cdot; 0, 0)\} + 2\hat{\varepsilon}_{\text{val}}(\delta) + 2\Delta_r. \end{aligned}$$

This does not give realized-sample universal dominance. Section 3.2.2 constructs candidate ratio estimators, including classifier, Riesz-representer, and balancing estimators. This corollary only propagates the drift of a supplied validation-independent $\hat{r}$; it neither constructs $\hat{r}$ nor establishes orthogonality with respect to the density-ratio direction.

The general fitting and validation results combine only after the fitting events hold simultaneously over the finite candidate set. The next theorem makes that requirement explicit.

Theorem 10 (Risk of the honestly selected general learner). Let $\hat{\tau}_1, \dots, \hat{\tau}_M$ be $M < \infty$ fitted candidates fixed before final validation and assume RCT consistency and conditional randomization, known strict-overlap trial propensity, honest nuisance fitting, independent RCT and OBS fitting samples, and final RCT and OBS validation samples independent of all candidate fitting. Suppose either the covariate marginals agree and $r_0 = 1$, or exact transport holds with $r_0 = dP_R^X/dP_O^X$. Allow independently trained ratios in fitting and validation and condition on them. For candidate k , let ζ_k° be its best-in-class target, let $u_k^\star \triangleq u_{k,n_f,N_f}^\star(\hat{\tau}_k; \delta_f/M)$, and define

$$B_k \triangleq \mathcal{R}(\zeta_k^\circ) + C_k\{u_k^\star + \Delta_k^{\text{fit}} + \text{Pen}_k(\zeta_k^\circ) + \epsilon_{\text{opt},k}\}.$$

Assume directly that $\mathcal{R}(\hat{\tau}_k) \leq B_k$ for every $k = 1, \dots, M$ simultaneously with probability at least $1 - \delta_f$. Let $\hat{k}$ be the deterministically tie-broken index selected by the common-reference criterion in Equation (67). Let $\hat{\varepsilon}_{\text{val}}(\delta_v)$ and $\Delta_r^{\text{val}} \triangleq \Delta_r$ be the validation concentration radius and validation ratio drift in Equation (70), respectively. Assume the validation-selection event

$$\mathcal{R}(\hat{\tau}_{\hat{k}}) \leq \min_{1 \leq k \leq M} \mathcal{R}(\hat{\tau}_k) + 2\hat{\varepsilon}_{\text{val}}(\delta_v) + 2\Delta_r^{\text{val}}$$

holds with probability at least $1 - \delta_v$. Then, with probability at least $1 - \delta_f - \delta_v$,

$$\mathcal{R}(\hat{\tau}_{\hat{k}}) \leq \min_{1 \leq k \leq M} B_k + 2\hat{\varepsilon}_{\text{val}}(\delta_v) + 2\Delta_r^{\text{val}}. \quad (71)$$

Here Δ_k^{fit} changes a fitted candidate, whereas Δ_r^{val} changes how fixed candidates are ranked. The result does not imply coefficient consistency, strict improvement, or approximation of the continuous coefficient oracle.

We next instantiate the abstract fitting bound and candidate construction in a class for which the estimator, covariance constants, and approximation rate are explicit.

4.5. Linear-sieve and additive-spline specialization

This specialization has two roles. It computes every fixed (λ, ω) candidate used in Section 4.4, and it turns the abstract two-source modulus into explicit covariance and approximation terms.

Definition 21 (Linear-sieve fusion estimator). Let $b(x) \in \mathbb{R}^p$ be fixed or honestly trained and define $\zeta_\beta(x) \triangleq b(x)^\top \beta$. Let $P = P^\top \succeq 0$ and $\rho \geq 0$. For $j \in \{\text{DRF}, \text{RF}\}$, define

$$\begin{aligned} \hat{J}_j(\beta; \lambda, \omega) &\triangleq \hat{L}_j(\zeta_\beta; \lambda, \omega) + \rho\beta^\top P\beta, \\ \hat{\beta}_{\lambda, \omega}^j &\in \arg \min_{\beta \in \mathbb{R}^p} \hat{J}_j(\beta; \lambda, \omega), \\ \hat{\tau}_j(x; \lambda, \omega) &\triangleq b(x)^\top \hat{\beta}_{\lambda, \omega}^j, \end{aligned}$$

using deterministic tie-breaking whenever needed.

For DRF, define

$$\begin{aligned}\hat{\Gamma}_R &\triangleq \mathbb{P}_R^{\text{fit}}(bb^\top), & \hat{\Gamma}_{O,r} &\triangleq \mathbb{P}_O^{\text{fit}}(r_0bb^\top), \\ \hat{h}_\lambda &\triangleq \mathbb{P}_R^{\text{fit}}(bZ_\lambda), & \hat{d}_{\text{OBS}} &\triangleq \mathbb{P}_O^{\text{fit}}(r_0b\hat{g}) - \mathbb{P}_R^{\text{fit}}(b\hat{g}).\end{aligned}$$

For RF, let

$$\begin{aligned}s_{m_\lambda} &\triangleq \frac{(A - e_0)(Y - m_\lambda)}{e_0(1 - e_0)}, & \hat{\Gamma}_{R,\kappa} &\triangleq \mathbb{P}_R^{\text{fit}}(\kappa bb^\top), \\ \hat{h}_{m_\lambda} &\triangleq \mathbb{P}_R^{\text{fit}}(bs_{m_\lambda}), & \hat{d}_{\text{OBS},\kappa} &\triangleq \mathbb{P}_O^{\text{fit}}(r_0b\hat{g}) - \mathbb{P}_R^{\text{fit}}(\kappa b\hat{g}).\end{aligned}$$

Proposition 18 (Normal equations and positive semidefiniteness). For DRF and RF, respectively, define

$$\begin{aligned}\hat{H}_{\text{DRF}}(\omega) &\triangleq (1 - \omega)\hat{\Gamma}_R + \omega\hat{\Gamma}_{O,r}, & \hat{h}_{\text{DRF}}(\lambda, \omega) &\triangleq \hat{h}_\lambda + \omega\hat{d}_{\text{OBS}}, \\ \hat{H}_{\text{RF}}(\omega) &\triangleq (1 - \omega)\hat{\Gamma}_{R,\kappa} + \omega\hat{\Gamma}_{O,r}, & \hat{h}_{\text{RF}}(\lambda, \omega) &\triangleq \hat{h}_{m_\lambda} + \omega\hat{d}_{\text{OBS},\kappa}.\end{aligned}$$

The normal equation is

$$\{\hat{H}_j(\omega) + \rho P\}\hat{\beta}_{\lambda,\omega}^j = \hat{h}_j(\lambda, \omega).$$

When the matrix is invertible,

$$\hat{\beta}_{\lambda,\omega}^{\text{DRF}} = \{\hat{H}_{\text{DRF}}(\omega) + \rho P\}^{-1}\hat{h}_{\text{DRF}}(\lambda, \omega), \quad (72)$$

$$\hat{\beta}_{\lambda,\omega}^{\text{RF}} = \{\hat{H}_{\text{RF}}(\omega) + \rho P\}^{-1}\hat{h}_{\text{RF}}(\lambda, \omega). \quad (73)$$

If $0 \leq \omega \leq 1$, the ratio weights are nonnegative, and $P \succeq 0$, both penalized Hessians are positive semidefinite. The signed source representation alone does not imply invertibility. A fit using $\hat{r}$ requires nonnegative weights, a direct Hessian check, or a prespecified positive-semidefinite projection.

For fixed ω , the Hessian does not depend on λ , while the right-hand side is affine. This yields an efficient candidate-generation device, not a replacement for final honest validation.

Proposition 19 (Fixed-scale profiling for candidate generation). Suppose $\hat{H}_j(\omega) + \rho P$ is invertible and write

$$\hat{h}_j(\lambda, \omega) \triangleq \hat{h}_{j,0}(\omega) + \lambda\hat{h}_{j,1}(\omega).$$

Then

$$\hat{\beta}_{\lambda,\omega}^j = \hat{\beta}_{0,\omega}^j + \lambda\hat{\beta}_\omega^j, \quad \hat{\beta}_\omega^j \triangleq \{\hat{H}_j(\omega) + \rho P\}^{-1}\hat{h}_{j,1}(\omega). \quad (74)$$

On a separate tuning sample, the corresponding quadratic criterion has the form

$$\hat{C}_j^{\text{tune}}(\lambda, \omega) = c_0(\omega) + 2c_1(\omega)\lambda + c_2(\omega)\lambda^2. \quad (75)$$

For a prespecified finite ω -grid, checking the admissible stationary point and the endpoints of $[0, 1]$ generates finitely many candidates. Those candidates must be fixed before the final validation used in Theorem 9. Continuous profiling on the final validation sample requires a separate uniform-convergence argument.

We now separate sieve approximation from estimation. Define

$$\begin{aligned}\Gamma_R &\triangleq \mathbb{E}_R(bb^\top), & \gamma_R &\triangleq \mathbb{E}_R(b\tau), \\ \beta_p &\triangleq \Gamma_R^{-1}\gamma_R, & \zeta_p &\triangleq b^\top \beta_p, & a_p^2 &\triangleq \mathbb{E}_R(\tau - \zeta_p)^2,\end{aligned}$$

and assume $\Gamma_R \succ 0$. For a fixed candidate, let $\hat{H}_j \triangleq \hat{H}_j(\omega)$, $\hat{h}_j \triangleq \hat{h}_j(\lambda, \omega)$, and define

$$\hat{u}_j \triangleq \hat{h}_j - \hat{H}_j \beta_p.$$

Theorem 11 (Exact random-Hessian sieve identity). For any realized fitting samples and fixed fitted nuisances, if $\hat{H}_j + \rho P$ is invertible, then

$$\hat{\beta}_j - \beta_p = (\hat{H}_j + \rho P)^{-1} \{\hat{u}_j - \rho P \beta_p\}, \quad (76)$$

$$\mathcal{R}(\hat{\tau}_j) = a_p^2 + (\hat{\beta}_j - \beta_p)^\top \Gamma_R (\hat{\beta}_j - \beta_p). \quad (77)$$

The identity is exact, but it is not a same-sample bias-variance decomposition because the random inverse Hessian and score are dependent.

For the concentration result, define the raw source contributions explicitly:

$$\begin{aligned}\psi_{R,\text{DRF}} &\triangleq b\{Z_\lambda - \omega \hat{g} - (1 - \omega)\zeta_p\}, & \psi_{O,\text{DRF}} &\triangleq \omega r_0 b(\hat{g} - \zeta_p), \\ \psi_{R,\text{RF}} &\triangleq b[s_{m_\lambda} - \kappa\{\omega \hat{g} + (1 - \omega)\zeta_p\}], & \psi_{O,\text{RF}} &\triangleq \omega r_0 b(\hat{g} - \zeta_p).\end{aligned}$$

Under the exact-target conditions, $\mathbb{E}_R \psi_{R,j} + \mathbb{E}_O \psi_{O,j} = 0$; the two source means need not vanish separately. Let

$$\Sigma_{R,j} \triangleq \text{Var}_R(\psi_{R,j}), \quad \Sigma_{O,j} \triangleq \text{Var}_O(\psi_{O,j}).$$

For $S \in \{R, O\}$, define

$$\nu_{S,j} \triangleq \|\Gamma_R^{-1/2} \Sigma_{S,j} \Gamma_R^{-1/2}\|_{\text{op}}.$$

Theorem 12 (Fixed-candidate covariance-sensitive bound). Assume RCT consistency and conditional randomization, known strict-overlap e_0 , exact r_0 , honestly fixed nuisances and basis, fixed $(j, \lambda, \omega, p, \rho)$, and independent fitting samples of sizes n_f, N_f . Assume the centered source vectors are sub-Gaussian with the displayed covariance proxies and the weighted designs satisfy, with failure probability at most δ_H ,

$$\mathcal{E}_H \triangleq \{\|\Gamma_R^{-1/2}(\hat{H}_j - \Gamma_R)\Gamma_R^{-1/2}\|_{\text{op}} \leq \eta\}, \quad 0 \leq \eta < 1.$$

Then, with probability at least $1 - \delta_H - \delta_S$,

$$\begin{aligned} \mathcal{R}(\hat{\tau}_j) \leq a_p^2 + \frac{C_{\text{sg}}}{(1-\eta)^2} & \left[\frac{\text{tr}(\Gamma_R^{-1}\Sigma_{R,j}) + \nu_{R,j} \log(4/\delta_S)}{n_f} \right. \\ & \left. + \frac{\text{tr}(\Gamma_R^{-1}\Sigma_{O,j}) + \nu_{O,j} \log(4/\delta_S)}{N_f} \right] + \frac{C_\rho \rho^2}{(1-\eta)^2} \|\Gamma_R^{-1/2} P \beta_p\|_2^2, \end{aligned} \quad (78)$$

where $\nu_{S,j}$ bounds the corresponding whitened covariance operator norm and C_{sg}, C_ρ are universal constants under the stated sub-Gaussian convention. Independence makes the source covariance proxies add; the bound has no artificial $1/\sqrt{n_f N_f}$ cross term.

The preceding theorem is a safe finite-sample upper bound. A trace expression for the leading risk requires a separate fixed-dimensional asymptotic result.

Proposition 20 (Fixed-dimensional leading expected risk). Assume RCT consistency and conditional randomization, known strict-overlap e_0 , honest fixed nuisances and basis, and either a common covariate marginal with $r_0 = 1$ or $P_R^X \ll P_O^X$ with exact $r_0 = dP_R^X/dP_O^X$. Fix (j, λ, ω, p) , and suppose the RCT and OBS fitting samples are independent with $n_f, N_f \rightarrow \infty$ at a stable sample-size ratio. Assume finite $(2 + \delta)$ -moments, $\Gamma_R \succ 0$, Gram convergence, and first-order negligible ridge. Define the linearization remainder

$$R_{n_f, N_f} \triangleq \hat{\beta}_j - \beta_p - \Gamma_R^{-1} [(\mathbb{P}_R - P_R)\psi_{R,j} + (\mathbb{P}_O - P_O)\psi_{O,j}],$$

and assume explicitly that

$$\mathbb{E}\|R_{n_f, N_f}\|_{\Gamma_R}^2 = o(n_f^{-1} + N_f^{-1}), \quad \|a\|_{\Gamma_R}^2 \triangleq a^\top \Gamma_R a.$$

Then

$$\hat{\beta}_j - \beta_p = \Gamma_R^{-1} [(\mathbb{P}_R - P_R)\psi_{R,j} + (\mathbb{P}_O - P_O)\psi_{O,j}] + R_{n_f, N_f},$$

and

$$\mathbb{E}\{\mathcal{R}(\hat{\tau}_j)\} - a_p^2 = \frac{\text{tr}(\Gamma_R^{-1}\Sigma_{R,j})}{n_f} + \frac{\text{tr}(\Gamma_R^{-1}\Sigma_{O,j})}{N_f} + o(n_f^{-1} + N_f^{-1}). \quad (79)$$

This is an asymptotic expected leading-risk expansion, not an exact finite-sample identity.

Corollary 3 (First-order RCT versus fusion comparison). Assume RCT consistency and conditional randomization, known strict-overlap e_0 , honest fixed nuisances and basis, and either a common covariate marginal with $r_0 = 1$ or $P_R^X \ll P_O^X$ with exact $r_0 = dP_R^X/dP_O^X$. Assume independent RCT and OBS fitting samples with $n_f, N_f \rightarrow \infty$ at a stable sample-size ratio. Consider a fixed fusion rule (λ, ω) and the strict RCT-only comparator $(\lambda, \omega) = (0, 0)$. The two rules use the same basis, penalty, RCT nuisance, and sample split. Assume finite $(2 + \delta)$ -moments, $\Gamma_R \succ 0$, Gram convergence, first-order negligible ridge, and for each rule the mean-square linearization remainder is $o(n_f^{-1} + N_f^{-1})$. Their difference in first-order expected

excess risk is

$$\mathcal{V}_{\text{FUS}}^{(1)} - \mathcal{V}_{\text{RCT}}^{(1)} \triangleq \frac{\text{tr}[\Gamma_R^{-1}\{\Sigma_{R,j}(\lambda, \omega) - \Sigma_{R,j}(0, 0)\}]}{n_f} + \frac{\text{tr}[\Gamma_R^{-1}\Sigma_{O,j}(\lambda, \omega)]}{N_f}. \quad (80)$$

The fixed fusion rule improves on the strict RCT-only comparator at first order if and only if the right-hand side of Equation (80) is negative. Equality does not order the higher-order remainders, and this comparison gives no universal finite-sample dominance.

An estimated ratio can perturb both the normal-equation right-hand side and its Hessian. Define

$$\hat{\Delta}_{h,j} \triangleq \omega \mathbb{P}_O^{\text{fit}}[(\hat{r} - r_0)b\hat{g}], \quad (81)$$

$$\hat{\Delta}_{H,j} \triangleq \omega \mathbb{P}_O^{\text{fit}}[(\hat{r} - r_0)bb^\top]. \quad (82)$$

Proposition 21 (Estimated-ratio perturbation of a fitted sieve). Condition on an independent ratio-training sample and use the same candidate specification with r_0 and $\hat{r}$. If both penalized Hessians are invertible, then

$$\hat{\beta}_j^{\hat{r}} - \hat{\beta}_j^{r_0} = \{\hat{H}_j^{\hat{r}} + \rho P\}^{-1}\{\hat{\Delta}_{h,j} - \hat{\Delta}_{H,j}\hat{\beta}_j^{r_0}\}.$$

Thus ratio normalization alone controls neither the score nor Hessian perturbation. This proposition instantiates the fitting drift in Proposition 14; it is not an orthogonality result.

Finally, the additive specialization exposes how the two sample sizes enter the rate through method-specific variance constants rather than a generic effective sample size.

Corollary 4 (Operational additive-spline rate). Suppose $\tau(x) = \sum_{\ell=1}^d \tau_\ell(x_\ell)$, with fixed d and smoothness $s > 0$. Let K denote the spline dimension and suppose $a_K^2 = O(K^{-2s})$. Assume RCT consistency and randomization, known strict-overlap e_0 , exact r_0 unless an explicit fitting-drift term is retained, honest fixed nuisances and basis, independent fitting samples, and uniformly bounded eigenvalues of Γ_R . Assume the centered source scores and weighted designs have uniformly bounded sub-Gaussian norms, $K + \log(1/\delta) = o\{\min(n_f, N_f)\}$, and the Gram event holds with probability tending to one. Define the per-dimension variance scale

$$c_{S,j}(K; \lambda, \omega) \triangleq K^{-1} \text{tr}\{\Gamma_R(K)^{-1}\Sigma_{S,j}^{(K)}(\lambda, \omega)\}, \quad S \in \{R, O\},$$

$$v_j(K; \lambda, \omega) \triangleq \frac{c_{R,j}(K; \lambda, \omega)}{n_f} + \frac{c_{O,j}(K; \lambda, \omega)}{N_f},$$

and suppress λ from $c_{O,j}$ only when the OBS score contribution is invariant to it. If the fitting ratio drift and ridge bias are $B_{r,j}^{\text{fit}}$ and $B_{\rho,j}$, then

$$\mathcal{R}(\hat{\tau}_j) = O_p\{K^{-2s} + K v_j(K; \lambda, \omega) + B_{r,j}^{\text{fit}} + B_{\rho,j}\}. \quad (83)$$

For a prespecified admissible dimension set $\mathcal{K}_n$, define the noncircular oracle dimension by

$$K_j^*(\lambda, \omega) \in \arg \min_{K \in \mathcal{K}_n} \{K^{-2s} + K v_j(K; \lambda, \omega)\}.$$

If, along the admissible sequence and uniformly over the coefficient candidates under consideration, $c_{S,j}(K; \lambda, \omega) \rightarrow c_{S,j}^\infty(\lambda, \omega)$, where the limiting constants are uniformly bounded away from zero and infinity, define

$$v_j^\infty(\lambda, \omega) \triangleq \frac{c_{R,j}^\infty(\lambda, \omega)}{n_f} + \frac{c_{O,j}^\infty(\lambda, \omega)}{N_f}.$$

Then a rate-optimal admissible choice satisfies $K \asymp \{v_j^\infty(\lambda, \omega)\}^{-1/(2s+1)}$, and the first two terms are of order $\{v_j^\infty(\lambda, \omega)\}^{2s/(2s+1)}$. For a validation-selected learner, the validation radius and validation ratio drift from Theorem 10 must also be added. The harmonic size $(1/n_f + 1/N_f)^{-1}$ is only a rough simplification when the two variance constants have comparable, uniformly bounded order.

4.6. Implementation boundaries and open theory

A fixed or honestly trained frozen neural representation can replace b , conditional on that representation. End-to-end neural fitting is nonconvex and is not covered. Estimated-ratio guarantees require uniform score, Hessian, and validation-loss control. Cross-fitting can recover data efficiency, but overlapping outer training sets do not justify a naive sum of fold variances. Neither target preservation nor candidate inclusion gives universal finite-sample improvement for estimated (λ, ω) .

5. Experiments

5.1. Completed ATE benchmarks

The completed experiments evaluate ATE estimation only. All nested benchmarks use 20 Monte Carlo repetitions per cell, an RCT-only AIPW baseline, a λ -only outcome-model fusion estimator, an ω -only PPI correction estimator, and the joint estimator. The code retains two stale internal names: `haipw_only` denotes the λ -only estimator and `ppi_power_tuned_only` denotes the ω -only estimator. The first label does not identify the method with the separate HAIPW literature.

The fixed-OBS-budget benchmark contains 16 design cells: three arbitrary nonlinear SCMs and one STAR real- X construction, each under a common covariate marginal and covariate shift and $n_R \in \{20, 30\}$, with $N_O = 5000$. The joint estimator has the smallest RMSE in 11 cells, and the λ -only estimator is best in the remaining five. The joint estimator has lower RMSE than AIPW in all 16 cells, with mean RMSE ratio 0.648 and mean empirical-variance ratio 0.439 relative to AIPW. The λ -only estimator also beats AIPW in all 16 cells, with mean RMSE ratio 0.679. The ω -only estimator is not best in any cell and has mean RMSE ratio 1.005.

The proportional-budget benchmark contains 32 cells with $n_R \in \{20, 30, 50, 100\}$ and $N_O = 100n_R$.

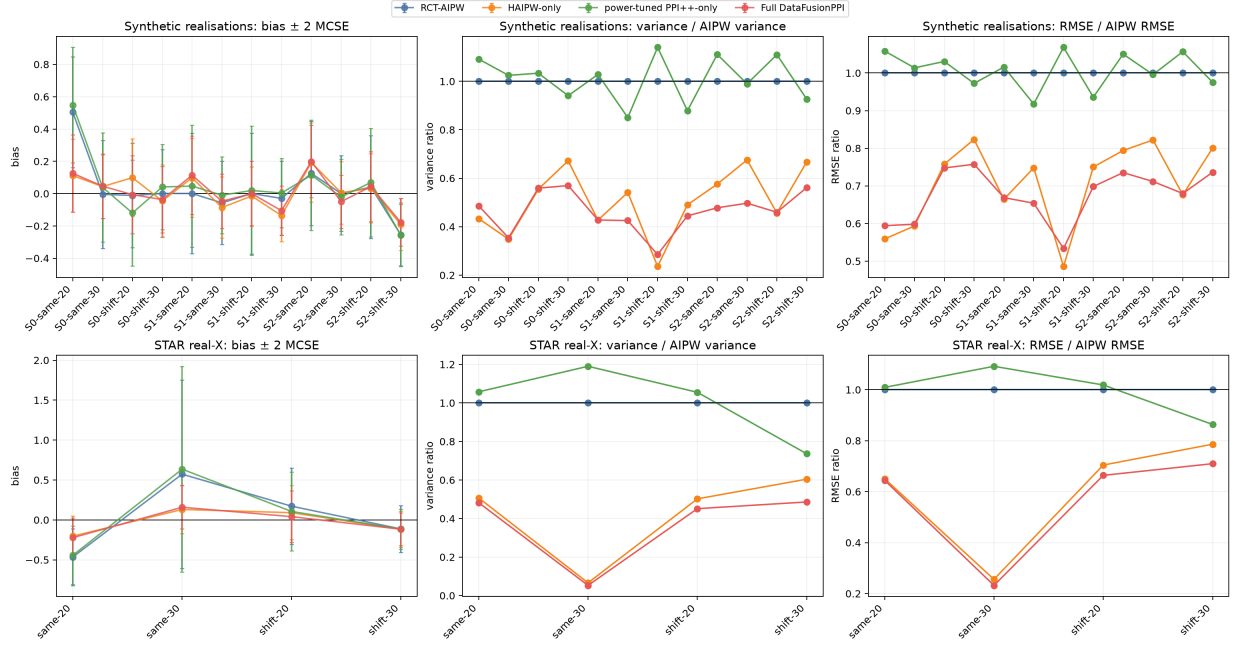


Figure 1: Fixed-OBS-budget ATE benchmark. The composite artifact reports multiple method and regime panels side by side for $n_R \in \{20, 30\}$ and $N_O = 5000$.

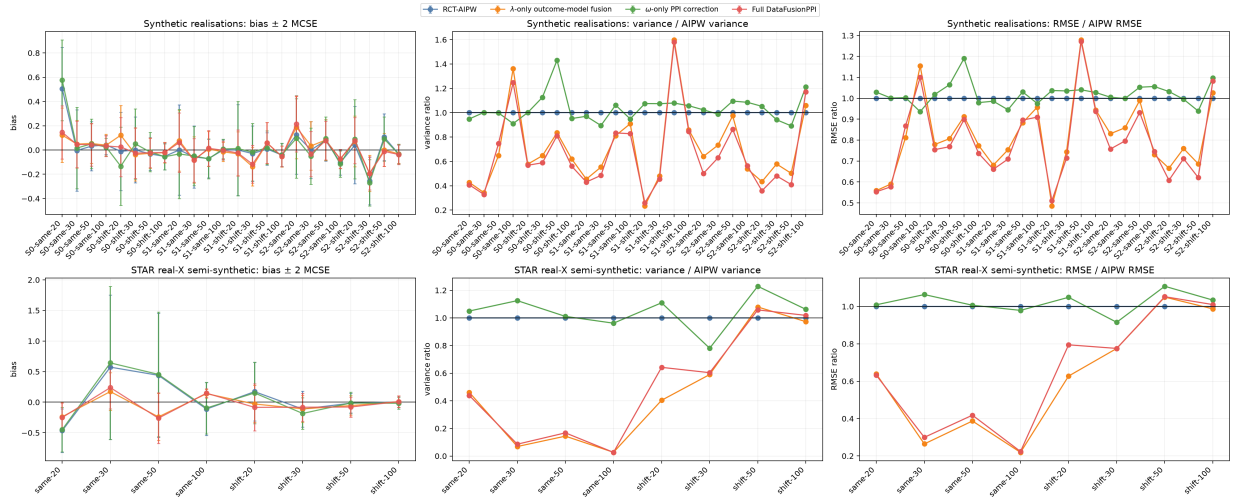


Figure 2: Proportional-budget ATE benchmark. The composite artifact reports multiple method and regime panels side by side for $n_R \in \{20, 30, 50, 100\}$ and $N_O = 100n_R$.

The joint estimator has the smallest RMSE in 18 cells, the λ -only estimator in 10, the ω -only estimator in one, and AIPW in three. The joint estimator has lower RMSE than AIPW in 27 of 32 cells, with mean RMSE ratio 0.760 and mean empirical-variance ratio 0.625. The λ -only estimator beats AIPW in 28 cells and has mean RMSE ratio 0.769. The ω -only estimator beats AIPW in 10 cells and has mean RMSE ratio 1.020. These counts show that joint tuning is often useful but is not uniformly best after coefficient estimation.

The STAR benchmark uses only real kindergarten covariates from Project STAR. Treatment assignments, latent confounding, potential outcomes, and effect heterogeneity are generated synthetically. It is therefore a real- X , semi-synthetic benchmark rather than an analysis of STAR’s original intervention or outcomes. The planned WHI CT+OS analysis has not been run because participant-level access, a semantic crosswalk, and a survival or censoring extension are still required.

5.2. Planned CATE evaluation and interpretation

CATE experiments are planned. The primary design will compare the proved AIPW-pseudo-outcome F-Learner with the prospective R-loss extension using additive cubic B-splines, frozen neural bases, and linear sieves under a common covariate marginal and covariate shift. Each study will compare trial-only, λ -only, ω -only, and joint tuning using held-out RCT-population CATE risk. Classifier-only, balancing-only, hybrid, and oracle transport weights will be separated. No CATE result is claimed from Figures 1 and 2.

The empirical conclusion is deliberately limited. The two-channel population oracle weakly improves the ATE variance objective when the trial-only coefficient is feasible, but estimated coefficients do not inherit a universal no-harm property. The present 20-repetition studies suggest that the λ channel is the more stable source of gain and that joint tuning can add value in some designs. More Monte Carlo repetitions, completed CATE experiments, additional real- X supports, and an authorized real-outcome analysis are required before making a stronger empirical claim.

5.3. Scope, failure modes, and conclusion

The framework uses observational information through prediction rather than by declaring the observational treatment contrast causal. Trial randomization validates the pseudo-outcome for every fixed outcome regression. A common covariate marginal or an exact density ratio centers the ATE control variate and the prediction-powered CATE loss, which creates separate λ - and ω -channels while retaining RCT-population targets.

The first boundary is covariate transport. Without weighting under $P_R^X \neq P_O^X$, the ATE correction has drift $\omega\{P_O\hat{g} - P_R\hat{g}\}$, and the unweighted CATE loss targets the wrong covariate average. Exact $r = dP_R^X/dP_O^X$ removes both discrepancies. An estimated ratio instead contributes the explicit ATE drift $\omega P_O\{(\hat{r} - r)\hat{g}\}$ and the CATE validation drift in Equation (70). Evaluation-only variance is first-order adequate only under the stated negligible-drift conditions; otherwise ratio-training influence and covariance terms are required.

The second boundary is trial propensity overlap. Proposition 1 uses the known, strictly positive randomized-trial propensity. Extreme inverse-propensity factors can make the required moments unstable even when the conditional-mean identity remains algebraically valid, and an incorrect propensity removes the exact arbitrary- q guarantee.

The third boundary is weak observational prediction. The unrestricted ω -gain depends on how strongly $\hat{g}(X)$ predicts $\tau(X)$, with attenuation from the finite n/N ratio. The λ -channel gives little gain when $\Delta = Z_1 - Z_0$ is uninformative or nearly degenerate. Population oracles diagnose these limits but do not make their empirical coefficient estimates stable.

The fourth boundary is adaptive reuse. Fitting nuisances, tuning candidates, and evaluating the selected candidate on the same outcomes invalidates the conditioning product law. Present inference therefore uses independent nuisance-fitting, coefficient-tuning, and final-evaluation samples, supplied separately or constructed by splitting. The theory does not cover cross-fitted reuse; that extension requires a separate stability or influence-function analysis.

The fifth boundary is learner restriction and curvature. Target preservation does not make a finite-sample restricted or highly penalized projection accurate. The square-completed representation is a nonnegative mixture only for $0 \leq \omega < 1$, and positive semidefiniteness at $\omega = 1$ must be read from the original quadratic. Coefficients outside $[0, 1]$, nonlinear parameterizations, and penalty rescaling require direct analysis of the original objective.

The sixth boundary is tail behavior. ATE variance identities require square-integrable scores; the Wald theorem additionally requires conditional Lindeberg control, a sufficient $(2 + \delta)$ -moment condition, and a nondegenerate variance limit. Fixed-loss variance tuning imposes fourth-moment-type requirements. Heavy tails can therefore destabilize oracle formulas even when their algebraic denominators are nonzero.

Within these boundaries, the randomized trial fixes the causal target, the OBS outcome model can reduce within- X pseudo-outcome noise through λ , and the OBS covariate sample can reduce predictable between- X variation through ω . The exact results are conditional identities and population-oracle comparisons. Empirical work must retain the identical trial-only fallback, report selection and denominator instability, and avoid interpreting them as a universal finite-sample no-harm guarantee.

References

- Angelopoulos, A. N., Bates, S., Fannjiang, C., Jordan, M. I., and Zrnic, T. (2023). Prediction-powered inference. *Science*, 382(6671):669–674.
- Angelopoulos, A. N., Duchi, J. C., and Zrnic, T. (2024). PPI++: Efficient prediction-powered inference. *Journal of Machine Learning Research*, 25(104):1–59.
- Bareinboim, E. and Pearl, J. (2016). Causal inference and the data-fusion problem. *Proceedings of the National Academy of Sciences*, 113(27):7345–7352.
- Cheng, D. and Cai, T. (2021). Adaptive combination of randomized and observational data. *Journal of the American Statistical Association*, 116(536):1863–1874.
- Chernozhukov, V., Newey, W. K., and Singh, R. (2021). De-biased machine learning of global and local parameters using regularized riesz representers.
- Chernozhukov, V., Newey, W. K., and Singh, R. (2022). Automatic debiased machine learning of causal and structural effects. *Econometrica*, 90(3):967–1027.
- Colnet, B., Mayer, I., Chen, G., Dieng, A., Li, R., Varoquaux, G., Vert, J.-P., Josse, J., and Yang, S. (2020). Causal inference methods for combining randomized trials and observational studies: a review. *arXiv preprint arXiv:2011.08047*.
- Foster, D. J. and Syrgkanis, V. (2023). Orthogonal statistical learning. *The Annals of Statistics*, 51(3):879–908.
- Hainmueller, J. (2012). Entropy balancing for causal effects: A multivariate reweighting method to produce balanced samples in observational studies. *Political Analysis*, 20(1):25–46.
- Imai, K. and Ratkovic, M. (2014). Covariate balancing propensity score. *Journal of the Royal Statistical Society: Series B*, 76(1):243–263.
- Kallus, N., Puli, A. M., and Shalit, U. (2018). Removing hidden confounding by experimental grounding. *Advances in Neural Information Processing Systems*, 31.
- Kennedy, E. H. (2023). Towards optimal doubly robust estimation of heterogeneous causal effects. *Electronic Journal of Statistics*, 17(2):3008–3049.
- Künzel, S. R., Sekhon, J. S., Bickel, P. J., and Yu, B. (2019). Metalearners for estimating heterogeneous treatment effects using machine learning. *Proceedings of the National Academy of Sciences*, 116(10):4156–4165.
- Li, X., Li, Y., Cui, Q., Li, L., and Zhou, J. (2022). Robust direct learning for causal data fusion. In *Proceedings of the 14th Asian Conference on Machine Learning*, volume 189 of *Proceedings of Machine Learning Research*.
- Nie, X. and Wager, S. (2021). Quasi-oracle estimation of heterogeneous treatment effects. *Biometrika*, 108(2):299–319.
- Rosenman, E., Basse, G., Owen, A., and Baiocchi, M. (2023). Combining observational and experimental datasets using shrinkage estimators. *Biometrics*, 79(4):2961–2973.
- Shi, X., Pan, Z., and Miao, W. (2023). Data integration in causal inference. *WIREs Computational*

Statistics, 15(1):e1581.

Yang, S., Gao, C., Zeng, D., and Wang, X. (2023). Elastic integrative analysis of randomised trial and real-world data for treatment heterogeneity estimation. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 85(3):575–596.

Yang, S., Liu, S., Zeng, D., and Wang, X. (2025). Data fusion methods for the heterogeneity of treatment effect and confounding function. *Bernoulli*, 31(4):2987–3012.

A. Omitted Preliminaries

A.1. Prediction-powered inference and PPI⁺⁺

Let (X_i, Y_i) , $i = 1, \dots, n$, be independent and identically distributed labeled observations from a distribution P , and let $\tilde{X}_j$, $j = 1, \dots, N$, be independent and identically distributed unlabeled covariates from the marginal distribution P^X , independently of the labeled sample. Let f be a fixed predictor. For a fixed coefficient $\omega \in \mathbb{R}$, the PPI⁺⁺ estimator ([Angelopoulos et al., 2024](#)) of the labeled-population mean is

$$\hat{\mu}_{\text{PPI}^{++}}(\omega) \triangleq \frac{1}{n} \sum_{i=1}^n Y_i + \omega \left\{ \frac{1}{N} \sum_{j=1}^N f(\tilde{X}_j) - \frac{1}{n} \sum_{i=1}^n f(X_i) \right\}. \quad (84)$$

If Y and $f(X)$ are integrable, then $\hat{\mu}_{\text{PPI}^{++}}(\omega)$ is unbiased for $\mathbb{E}(Y)$ for every fixed ω , because the labeled and unlabeled predictions have the same expectation. The same conclusion holds conditionally when ω is selected using data independent of the two samples displayed above. PPI⁺⁺ selects ω to control how strongly this mean-zero prediction difference enters the estimator. When $\text{Cov}\{Y, f(X)\} \neq 0$ and ω is close to its variance-minimizing value, the prediction difference can reduce variance relative to the labeled-only estimator.

B. Simulation Details

All completed simulation artifacts were generated with base seed 190602. The primary nested benchmarks use 20 outer Monte Carlo repetitions per design cell and compare four estimators: RCT-only AIPW, λ -only outcome-model fusion, ω -only PPI correction, and joint (λ, ω) fusion. The implementation retains the historical internal codes `haipw_only` and `ppi_power_tuned_only` for the second and third estimators, respectively. These codes are not the user-facing method names, and the first does not refer to the separate HAIPW paper.

The arbitrary simulation family uses three nonlinear structural causal models. In every model, a domain indicator S distinguishes the observational sample ($S = 0$) from the randomized sample ($S = 1$); covariates depend on latent variables, observational treatment depends on both measured covariates and an unmeasured confounder, trial treatment follows the known randomized propensity, and outcomes may depend nonlinearly on covariates, treatment, and the latent confounder. The common-marginal regime uses $P_R^X = P_O^X$. The shifted regime changes the covariate distribution while preserving overlap and estimates the density ratio from a domain classifier. The same estimators and evaluation logic are used in both regimes.

The fixed-OBS-budget experiment has 16 cells: four benchmark families, two covariate regimes, and $n_R \in \{20, 30\}$, with $N_O = 5000$. The proportional experiment has 32 cells: four benchmark families, two regimes, and $n_R \in \{20, 30, 50, 100\}$, with the exact mapping $N_O = 100n_R \in \{2000, 3000, 5000, 10000\}$. Coefficients are estimated by the nested honest procedure; no oracle coefficient or oracle density ratio is used in the reported method comparison. The replication-level CSV files, summaries, and composite plots in `materials/` are the canonical numerical artifacts.

The fourth benchmark family reuses the empirical covariate support from the Tennessee Student/Teacher Achievement Ratio (STAR) experiment. Treatment, latent confounding, outcomes, and treatment-effect heterogeneity are generated by the benchmark code. Consequently this is a real- X , semi-synthetic design, not an evaluation of the original STAR outcomes or intervention effect.

Figure 3 records the earlier two-method pilot. It is retained for provenance but is not the basis for the four-method conclusions in Section 5.

No participant-level WHI CT+OS analysis is included. Such an analysis requires authorized data access, a verified trial-observational semantic crosswalk, a prespecified estimand, and methods appropriate for censoring if the selected outcome is time-to-event. No completed CATE experiment or real-outcome result is represented by the present artifacts.

Illustrative random-SCM debugging pilot (20 outer refits per cell)
per stage: RCT n / OBS 5000; total generated per replicate: $3n / 15000$

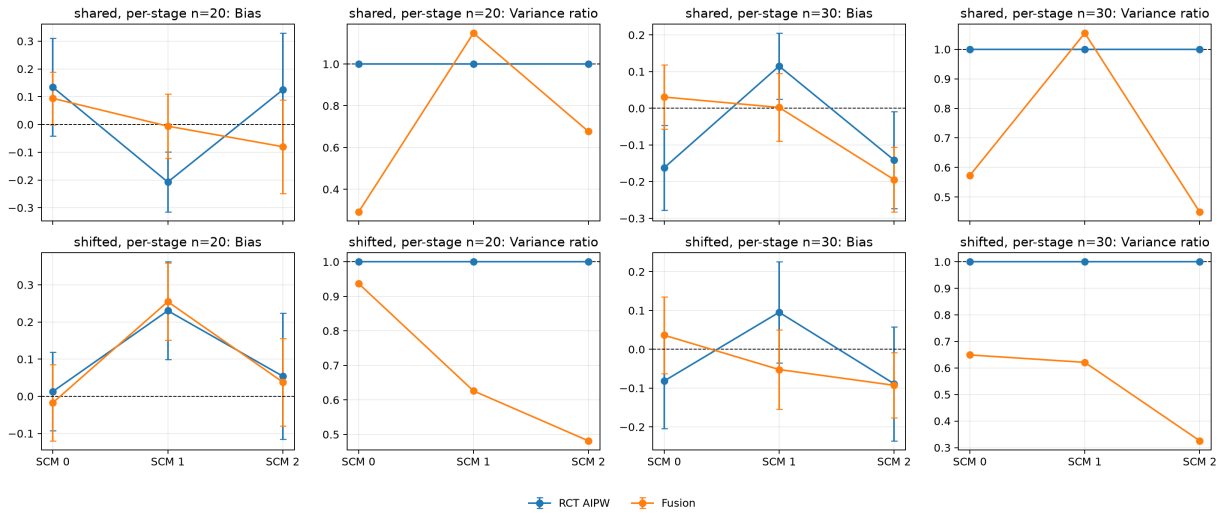


Figure 3: Earlier SSEM ATE pilot. This artifact predates the nested four-method comparison and should be interpreted only as a design and implementation check.

C. Proof Details

Unless stated otherwise, all proofs use Assumption 1: every nuisance, ratio, candidate function, and coefficient is fixed before the independent RCT and OBS evaluation observations are used.

C.1. Conditional validity of the trial pseudo-outcome

Fix $q = (q_0, q_1)$ without using the evaluation samples and condition on X . Under trial randomization, consistency, and honest sample separation,

$$\mathbb{E}_R \left[\frac{A}{e(X)} \{Y - q_1(X)\} \mid X \right] = \mathbb{E}_R \{Y(1) - q_1(X) \mid X\}, \quad (85)$$

$$\mathbb{E}_R \left[\frac{1 - A}{1 - e(X)} \{Y - q_0(X)\} \mid X \right] = \mathbb{E}_R \{Y(0) - q_0(X) \mid X\}. \quad (86)$$

Substituting these two equalities into the definition of φ cancels the two fitted regressions and yields $\mathbb{E}_R \{\varphi(V; q) \mid X\} = \tau(X)$. Applying the identity to $q = \hat{\mu}_R$, $q = \hat{\mu}_O$, and their affine blend gives

$$\mathbb{E}_R(Z_\lambda \mid X) = \tau(X), \quad \mathbb{E}_R(\Delta \mid X) = 0. \quad (87)$$

C.2. Proof of Corollary 1

Apply Equation (6) with the nuisance-sample fitted pair $q = \hat{\mu}_R$, average over D_R^{eval} , and use iterated expectation. This gives

$$\mathbb{E}\{\hat{\theta}_{\text{AIPW}, R}\} = \mathbb{E}_R\{\tau(X)\} = \theta.$$

C.3. Proof of Theorem 1

Pseudo-outcome validity and exact transport give

$$P_R(Z_\lambda) = \theta, \quad P_O(r_0 \hat{g}) = P_R(\hat{g}).$$

Substituting these identities into Equation (13) proves Equation (14). For any fixed $\tilde{r}$,

$$\begin{aligned} \mathbb{E}\{\hat{\theta}_{\tilde{r}}(\lambda, \omega)\} - \theta &= \omega \{P_O(\tilde{r} \hat{g}) - P_R(\hat{g})\} \\ &= \omega P_O\{(\tilde{r} - r_0) \hat{g}\}, \end{aligned}$$

which proves Equation (15). Using the same pseudo-outcome and transport identities, rewrite the centered estimator as

$$\hat{\theta}_{r_0}(\lambda, \omega) - \theta = (\mathbb{P}_{R,n} - P_R)\{Z_\lambda - \omega \hat{g}\} + (\mathbb{P}_{O,N} - P_O)\{\omega r_0 \hat{g}\}. \quad (88)$$

The two summands are independent because D_R^{eval} and D_O^{eval} are independent. The variance of an empirical mean of m independent observations is the population variance divided by m , which proves Equation (16). Unbiased sample variances on the two independent evaluation blocks therefore give the displayed fixed-selected-coefficient variance estimator in Section 3. Coverage of its

Wald interval additionally uses the stated evaluation-sample Lindeberg and moment assumptions; variance unbiasedness alone is not a central limit theorem.

C.4. Proof of Corollary 1.1

Expand the first variance in Equation (16) using $Z_\lambda = Z_0 + \lambda\Delta$. The interaction coefficient vanishes because

$$\begin{aligned}\text{Cov}_R(\Delta, \hat{g}) &= \mathbb{E}_R[\{\hat{g} - \mathbb{E}_R(\hat{g})\}\mathbb{E}_R(\Delta \mid X)] = 0, \\ D &= \text{Cov}_R(Z_0, \hat{g}) = \text{Cov}_R\{\mathbb{E}_R(Z_0 \mid X), \hat{g}\} = \text{Cov}_R\{\tau(X), \hat{g}(X)\}.\end{aligned}\quad (89)$$

Collecting the coefficient-dependent terms and multiplying by n gives $2C\lambda - 2D\omega + A\lambda^2 + B\omega^2$, which proves Equation (18). Finally, for every (u, v) ,

$$\begin{pmatrix} u & v \end{pmatrix} \mathbf{H}_{\text{ATE}} \begin{pmatrix} u \\ v \end{pmatrix} = Au^2 + Bv^2 \geq 0,$$

so $\mathbf{H}_{\text{ATE}}$ is positive semidefinite.

C.5. Proof of Proposition 2

Equation (18) is the positive affine transformation

$$\text{Var}\{\hat{\theta}_{r_0}(\lambda, \omega)\} = \frac{1}{n} \text{Var}_R(Z_0) + \frac{1}{n} \mathcal{J}_{\text{ATE}}(\lambda, \omega),$$

where $n > 0$. Adding a constant and multiplying by a positive scalar preserve the complete argmin set over every feasible set.

C.6. Proof of Theorem 2

When $A, B > 0$, the two first-order conditions are

$$2C + 2A\lambda = 0, \quad -2D + 2B\omega = 0.$$

The separated quadratic is strictly convex, so their unique solution is $\lambda^* = -C/A$ and $\omega^* = D/B$. For every $\rho \geq 0$, the ridge objective has gradient

$$\begin{pmatrix} 2C + 2(A + \rho)\lambda \\ -2D + 2(B + \rho)\omega \end{pmatrix}.$$

Since $A, B > 0$, both $A + \rho$ and $B + \rho$ are positive for every $\rho \geq 0$. Setting the gradient to zero therefore proves Equation (19) and uniqueness. At $\rho = 0$, the penalized objective reduces to $\mathcal{J}_{\text{ATE}}$, whose minimizers are also the minimizers of the exact population variance by Proposition 2.

C.7. Compact constraints

Continuity of $\mathcal{J}_{\text{ATE}}$ guarantees a minimizer on every nonempty compact rectangle. If the rectangle contains the origin, its minimum variance cannot exceed the RCT-only variance.

C.8. Proof of Corollary 2.1

For $A, B > 0$, complete the two scalar squares:

$$\mathcal{J}_{\text{ATE}}(\lambda, \omega) = A \left(\lambda + \frac{C}{A} \right)^2 + B \left(\omega - \frac{D}{B} \right)^2 - \frac{C^2}{A} - \frac{D^2}{B}.$$

The squared terms vanish at the oracle coefficients in Equation (20). Substitution into Equation (18) proves Equation (22).

C.9. Prediction-powered CATE losses and finite-sample theory

Fix $t, \lambda, \omega, \hat{g}, r_0$ before using the fitting sample. Conditional on X , pseudo-outcome validity gives $\mathbb{E}_R(Z_\lambda - \tau(X) \mid X) = 0$, and hence

$$\mathbb{E}_R\{Z_\lambda - t(X)\}^2 = \mathbb{E}_R\{Z_\lambda - \tau(X)\}^2 + \mathbb{E}_R\{t(X) - \tau(X)\}^2. \quad (90)$$

Exact transport also gives

$$\mathbb{E}_O\{r_0(X)(\hat{g}(X) - t(X))^2\} = \mathbb{E}_R\{\hat{g}(X) - t(X)\}^2. \quad (91)$$

The correction in Equation (50) therefore has mean zero, which proves Proposition 10.

For the RF identity, expand conditionally on X . Randomization implies

$$\begin{aligned} \mathbb{E}_R[(A - e_0)\{Y - m(X)\} \mid X] &= e_0(X)\{1 - e_0(X)\}\tau(X), \\ \mathbb{E}_R[\{A - e_0(X)\}^2 \mid X] &= e_0(X)\{1 - e_0(X)\}. \end{aligned}$$

It follows that

$$\mathcal{L}_R(\zeta; m) = Q_R(m) + \mathcal{R}(\zeta).$$

Because $\mathbb{E}_R(\kappa \mid X) = 1$ and exact transport holds, the RF correction has mean zero. This proves Theorem 6.

C.10. Proofs for general two-source learning

For Theorem 7, exact transport and the target identities in Section 4.1 imply

$$L_j(\zeta) - L_j(\zeta_j^\circ) = \mathcal{R}(\zeta) - \mathcal{R}(\zeta_j^\circ).$$

The global approximate-ERM inequality gives

$$\widehat{L}_j(\widehat{\zeta}_j) + \text{Pen}(\widehat{\zeta}_j) \leq \widehat{L}_j(\zeta_j^\circ) + \text{Pen}(\zeta_j^\circ) + \epsilon_{\text{opt}}.$$

Add and subtract the two population losses. By the definition of the centered comparator process, the resulting stochastic term is exactly $-\mathbb{Z}_j(\widehat{\zeta}_j)$. This proves Equation (61), including its sign and without an extra factor two.

For Theorem 8, work on the intersection of the two sourcewise modulus events. Independence and a union bound give probability at least $1 - 2\delta$. The exact ERM reduction shows that any approximate minimizer with excess risk in a peeling shell $2^m u < \mathcal{R}(\zeta) - \mathcal{R}(\zeta_j^\circ) \leq 2^{m+1} u$ must satisfy an upper bound by the sum of the two source moduli, the penalty at the comparator, and ϵ_{opt} . Sub-root regularity and the definition of u_{j,n_f,N_f}^* make that upper bound strictly smaller than the shell's lower endpoint once u is a sufficiently large universal multiple of $u_{j,n_f,N_f}^* + \text{Pen}(\zeta_j^\circ) + \epsilon_{\text{opt}}$. For each fixed δ , the full-modulus sub-root condition gives $\Phi_{j,n_f,N_f}(tu; \delta) \leq \sqrt{t} \Phi_{j,n_f,N_f}(u; \delta)$ for every $t \geq 1$. Thus the stochastic upper bound grows only as the square root of the peeling level, whereas the shell's excess-risk lower endpoint grows linearly. Summing the geometric tail allocation over shells proves Equation (62). Separability permits the displayed suprema to be taken over countable dense subclasses; otherwise the same argument is understood in outer probability.

For Proposition 14, condition on the ratio-training sample. The preceding argument now uses the actual $\widehat{r}$ -weighted source process, hence its fixed point is $u_{j,n_f,N_f}^*(\widehat{r}; \delta)$. Add and subtract the exact-ratio population loss at ζ and ζ_j° . Their increment is bounded by Δ_j^{fit} , after which the same localization argument closes at $\bar{u}$. If the two stated $L^2(P_O)$ bounds hold, Cauchy–Schwarz gives

$$\Delta_j^{\text{fit}}(u; \widehat{r}) \leq |\omega| \|\widehat{r} - r_0\|_{L^2(P_O)} \sup_{\zeta \in \mathcal{Z}_j(u)} \|G_\zeta - G_{\zeta_j^\circ}\|_{L^2(P_O)},$$

which proves the final claim.

C.11. Proofs for coefficient selection

For Proposition 15, pseudo-outcome validity gives $\mathbb{E}_R(\Delta \mid X) = 0$. Therefore

$$\begin{aligned} Q_{\text{DR}}(\lambda) &= Q_{\text{DR}}(0) + 2\lambda \mathbb{E}_R[(Z_0 - \tau)\Delta] + \lambda^2 \mathbb{E}_R(\Delta^2), \\ \mathbb{E}_R[(Z_0 - \tau)\Delta] &= \mathbb{E}_R(Z_0 \Delta). \end{aligned}$$

Completing the square gives Equation (65) when $\mathbb{E}_R(\Delta^2) > 0$. If $\mathbb{E}_R(\Delta^2) = 0$, then $\Delta = 0$ almost surely and the objective is flat.

For the R loss, write $s_0 \triangleq e_0(1 - e_0)$ and $U_R \triangleq Y - m_R - (A - e_0)\tau$. Since $m_\lambda = m_R + \lambda d_m$,

$$\begin{aligned} Q_R(m_\lambda) &= Q_R(m_R) - 2\lambda \mathbb{E}_R(U_R d_m / s_0) + \lambda^2 \mathbb{E}_R(d_m^2 / s_0), \\ \mathbb{E}_R(U_R d_m / s_0) &= \mathbb{E}_R[\{Y - m_R\} d_m / s_0]. \end{aligned}$$

The second identity follows from $\mathbb{E}_R(A - e_0 \mid X) = 0$. Completing the square proves Equation (66). If $\mathbb{E}_R(d_m^2 / s_0) = 0$, strict overlap implies $d_m = 0$ almost surely and the objective is flat. A convex quadratic minimized over $[0, 1]$ is minimized by projecting its unrestricted minimizer onto that interval.

For Proposition 16, exact transport gives

$$\mathbb{E}_O(r_0 G_\zeta) - \mathbb{E}_R(G_\zeta) = 0.$$

Moreover, conditional randomization and strict overlap give $\mathbb{E}_R(\kappa \mid X) = 1$, so

$$\begin{aligned}\mathbb{E}_R(\kappa G_\zeta) &= \mathbb{E}_R[\mathbb{E}_R(\kappa \mid X) G_\zeta] = \mathbb{E}_R(G_\zeta), \\ \mathbb{E}_O(r_0 G_\zeta) - \mathbb{E}_R(\kappa G_\zeta) &= 0.\end{aligned}$$

Substitution into Equations (50) and (52) leaves $\mathcal{L}_{\text{DR}}(\zeta; \lambda)$ and $\mathcal{L}_R(\zeta; m_\lambda)$, respectively, for every $\omega \in \Omega$. Each objective is therefore constant on the nonempty set Ω , and its minimizer set is all of Ω .

For Proposition 17, condition on the fitted grid function at each $(\lambda, \omega) \in \mathcal{G}_j$. Conditional pseudo-outcome validity gives

$$\mathbb{E}_R\{Z_0 - \hat{\tau}_j(X; \lambda, \omega)\}^2 = Q_{\text{DR}}(0) + \mathcal{R}\{\hat{\tau}_j(\cdot; \lambda, \omega)\}.$$

Exact transport gives $\mathbb{E}_O(r_0 G_{j, \lambda, \omega}) = \mathbb{E}_R(G_{j, \lambda, \omega})$, so the term multiplied by ω has mean zero. This proves Equation (68).

C.12. Proofs for the orthogonality audit

Condition throughout on the auxiliary samples and algorithmic randomness, so all nuisances, λ , and ω are fixed before the independent loss observations are used. Assume strict trial overlap, exact transport for claims at r_0 , and sufficient integrability and domination to interchange Gateaux differentiation and expectation. Directions belong to the tangent sets stated in the main text.

Differentiating Equations (50) and (52) in the target direction v gives Equations (55) and (56). For $r = r_0$, exact transport gives $\mathbb{E}_O(r_0 f) = \mathbb{E}_R(f)$ for each integrable $f(X)$. Also,

$$\mathbb{E}_R(\kappa_0 \mid X) = \mathbb{E}_R(\kappa \mid X) = \frac{\mathbb{E}_R[\{A - e_0(X)\}^2 \mid X]}{s_0(X)} = 1.$$

The two correction terms in each learning equation therefore cancel. This proves Proposition 11.

For the q -direction in Proposition 12, let $q_\varepsilon = q + \varepsilon h$. Direct differentiation gives

$$\dot{Z}_q[h] = -\{A - e_0(X)\} \left\{ \frac{h_1(X)}{e_0(X)} + \frac{h_0(X)}{1 - e_0(X)} \right\}.$$

Because h and v are functions of X and $\mathbb{E}_R(A - e_0 \mid X) = 0$, $\mathbb{E}_R\{v(X) \dot{Z}_q[h]\} = 0$.

For the m -direction, direct differentiation of the first term of Equation (56) gives

$$D_m \Psi_{\text{RF}}[h, v] = -\mathbb{E}_R \left[v(X) \frac{A - e_0(X)}{s_0(X)} h(X) \right] = 0.$$

The last equality again uses conditional randomization.

For the prediction path $\hat{g}_\varepsilon = \hat{g} + \varepsilon u$, only the transported correction changes. Exact transport, and

$\mathbb{E}_R(\kappa_0 \mid X) = 1$ for RF, give

$$\begin{aligned} \left. \frac{d}{d\varepsilon} \Psi_j(\hat{g}_\varepsilon)[v] \right|_{\varepsilon=0} &= \omega \mathbb{E}_O[\{r - r_0\}uv], \\ \left. \frac{d}{d\varepsilon} \mathbb{E}\{\hat{L}_j(\hat{g}_\varepsilon)\} \right|_{\varepsilon=0} &= 2\omega \mathbb{E}_O[\{r - r_0\}(\hat{g} - \zeta)u], \end{aligned}$$

which proves Equations (57) and (58). A ratio path $r_\varepsilon = r + \varepsilon h_r$ affects only the positive OBS term, so

$$\begin{aligned} D_r \Psi_j[h_r, v] &= \omega \mathbb{E}_O[h_r v (\hat{g} - \zeta)], \\ D_r \mathbb{E}(\hat{L}_j)[h_r] &= \omega \mathbb{E}_O[h_r (\hat{g} - \zeta)^2]. \end{aligned}$$

This proves Equations (59) and (60).

To prove Proposition 13, take any $f = v(\hat{g} - \zeta) \in L^2(P_O^X)$. Its inner product with every element of $L_0^2(P_O^X)$ is zero if and only if f lies in the orthogonal complement of that space, which consists of the almost-sure constants. For example, the centered direction $h = f - \mathbb{E}_O(f)$ has pairing $\mathbb{E}_O(hf) = \text{Var}_O(f)$. Multiplication by nonzero ω does not change the zero set.

For Proposition 18, expand each quadratic objective in β . The DRF curvature and right-hand side are $(1 - \omega)\hat{\Gamma}_R + \omega\hat{\Gamma}_{O,r} + \rho P$ and $\hat{h}_\lambda + \omega\hat{d}_{\text{OBS}}$. The RF counterparts are $(1 - \omega)\hat{\Gamma}_{R,\kappa} + \omega\hat{\Gamma}_{O,r} + \rho P$ and $\hat{h}_{m_\lambda} + \omega\hat{d}_{\text{OBS},\kappa}$. Setting the gradients to zero proves Equations (72) and (73). For $0 \leq \omega \leq 1$, each unpenalized Hessian is a convex combination of positive-semidefinite weighted Gram matrices; adding $\rho P \succeq 0$ preserves positive semidefiniteness.

For Proposition 19, fixed ω makes the Hessian independent of λ , while $Z_\lambda = Z_0 + \lambda\Delta$ and $m_\lambda = m_R + \lambda d_m$ make the right-hand side affine. Multiplication by the fixed inverse Hessian proves Equation (74). Substitution of the affine fitted function into the separate tuning criterion and expansion of its squares proves Equation (75). The coefficients are the constant, linear, and quadratic terms of that expansion.

For Theorem 9, Hoeffding's inequality and a union bound over the $M = |\mathcal{G}_j|$ grid pairs gives the uniform deviation ε_{val} . Comparing the selected score with the grid-oracle score and cancelling the common $Q_{\text{DR}}(0)$ gives the factor two.

For Corollary 2, direct expansion with the evaluation-independent approximate ratio perturbs the expectation at each grid pair by $\omega \mathbb{E}_O[(\hat{r} - r_0)G_{j,\lambda,\omega}]$. The main-text quantity Δ_r bounds these perturbations uniformly, so the same selector comparison adds $2\Delta_r$. Hoeffding's inequality applied to the actual scores gives $2\hat{\varepsilon}_{\text{val}}(\delta)$, proving both displayed bounds.

For Theorem 10, assign fitting failure probability δ_f/M to each of the M fixed candidates. A union bound makes all candidate-specific fitting inequalities hold simultaneously with probability at least $1 - \delta_f$. On that event, the ideal risk of candidate k is at most B_k . Independently, the validation event in Corollary 2 holds with probability at least $1 - \delta_v$ and compares the selected fitted candidate with the best fitted candidate, adding $2\hat{\varepsilon}_{\text{val}}(\delta_v) + 2\Delta_r^{\text{val}}$. Intersecting the fitting and validation events and then minimizing the simultaneous upper bounds over k proves Equation (71). No independence among the candidate fits is required; only their joint independence from final validation is used.

For Theorem 11, the first-order condition is $(\hat{H}_j + \rho P)\hat{\beta}_j = \hat{h}_j$. Subtracting its value at β_p proves

Equation (76). Projection orthogonality, $\mathbb{E}_R[b(\zeta_p - \tau)] = 0$, removes the risk cross term and proves Equation (77).

For Theorem 12, let $\psi_{R,j}$ and $\psi_{O,j}$ be the explicit raw contributions in the main text. Their source means need not vanish separately, but exact transport gives $P_R\psi_{R,j} + P_O\psi_{O,j} = 0$. Center each source by its own mean. Since the centered RCT and OBS sample means are independent, their covariance proxies add:

$$\text{Cov}(\mathbb{P}_R\psi_{R,j} + \mathbb{P}_O\psi_{O,j}) = \frac{\Sigma_{R,j}}{n_f} + \frac{\Sigma_{O,j}}{N_f}.$$

The stated sub-Gaussian vector concentration inequality bounds the squared $\Gamma_R^{-1/2}$ -norm by the sum of the two trace terms and their operator norm tail terms, without applying a triangle inequality between sources. On $\mathcal{E}_H$, the whitened inverse penalized Hessian has norm at most $(1 - \eta)^{-1}$. Combining this fact with Equation (76) and bounding the ridge vector separately proves Equation (78).

For Proposition 20, write the leading term as $A_{n_f, N_f} \triangleq \Gamma_R^{-1}\{(\mathbb{P}_R - P_R)\psi_{R,j} + (\mathbb{P}_O - P_O)\psi_{O,j}\}$, so the remainder defined in the proposition gives $\hat{\beta}_j - \beta_p = A_{n_f, N_f} + R_{n_f, N_f}$. Insert this identity into Equation (77). Independence removes the cross covariance between the RCT and OBS components of A_{n_f, N_f} , and hence

$$\mathbb{E}\|A_{n_f, N_f}\|_{\Gamma_R}^2 = \frac{\text{tr}(\Gamma_R^{-1}\Sigma_{R,j})}{n_f} + \frac{\text{tr}(\Gamma_R^{-1}\Sigma_{O,j})}{N_f}.$$

Moreover, Cauchy–Schwarz and the stated mean-square remainder condition give

$$\left| \mathbb{E}\langle A_{n_f, N_f}, R_{n_f, N_f} \rangle_{\Gamma_R} \right| \leq \{\mathbb{E}\|A_{n_f, N_f}\|_{\Gamma_R}^2\}^{1/2} \{\mathbb{E}\|R_{n_f, N_f}\|_{\Gamma_R}^2\}^{1/2} = o(n_f^{-1} + N_f^{-1}),$$

and the expected squared remainder has the same smaller order. The identities $\text{tr}(AB) = \text{tr}(BA)$ then give Equation (79). The comparison in Corollary 3 follows by subtracting the two leading expansions; it is not a finite-sample comparison.

For Proposition 21, write the estimated- ratio normal equation as $(\hat{H}_j^{r_0} + \hat{\Delta}_{H,j} + \rho P)\hat{\beta}_j^r = \hat{h}_j^{r_0} + \hat{\Delta}_{h,j}$. Subtract the exact-ratio equation and rearrange. Invertibility gives the displayed resolvent identity.

Finally, substitute $a_K^2 = O(K^{-2s})$ and the covariance scales

$$\frac{Kc_{R,j}(K; \lambda, \omega)}{n_f} \quad \text{and} \quad \frac{Kc_{O,j}(K; \lambda, \omega)}{N_f}$$

into the finite-sample result. Adding the stated ridge and fitting-ratio terms proves Equation (83). Minimizing the resulting criterion over $\mathcal{K}_n$ gives the displayed definition of K_j^* . Under the stated uniform positive finite limits, that criterion is asymptotically equivalent, uniformly over the coefficient candidates, to $K^{-2s} + Kv_j^\infty$. Balancing its two terms gives $K \asymp (v_j^\infty)^{-1/(2s+1)}$ and rate $(v_j^\infty)^{2s/(2s+1)}$, proving Corollary 4.

C.13. A two-stratum oracle calculation

Let $P_R(X = 0) = P_R(X = 1) = 1/2$, $\tau(0) = 0$, $\tau(1) = 2$, and $\hat{g} = \tau$. With $n = 100$, $N = 900$, and a common covariate marginal,

$$\mathbb{E}_R\{\tau(X)\} = 1, \quad \text{Var}_R\{\tau(X)\} = 1, \quad \omega_{\text{ATE}}^* = \frac{1}{1 + 100/900} = 0.9.$$

The baseline between- X variance is 0.01, whereas the oracle component is

$$\frac{\text{Var}_R\{0.1\tau(X)\}}{100} + \frac{0.9^2 \text{Var}_R\{\tau(X)\}}{900} = 0.0001 + 0.0009 = 0.001.$$

The reduction 0.009 concerns only between- X variation. The first RCT variance term in Equation (16) also contains conditional pseudo-outcome noise.

C.14. Honest variance estimation and evaluation-sample inference

For Proposition 3, fix the coefficients and scores before using the evaluation observations. Ordinary unbiased sample-variance identities give

$$\mathbb{E}\{\widehat{\text{Var}}_{R,\text{eval}}(\psi_R)\} = \text{Var}_R(\psi_R), \quad \mathbb{E}\{\widehat{\text{Var}}_{O,\text{eval}}(\psi_O)\} = \text{Var}_O(\psi_O).$$

Dividing by n and N , then adding, proves unbiasedness of $\hat{V}$ by independence of D_R^{eval} and D_O^{eval} .

For Theorem 4, impose the following evaluation regularity conditions after fixing the learned objects:

- (R1) the RCT and OBS evaluation observations are conditionally independent and retain their respective product laws;
- (R2) $n, N \rightarrow \infty$, $n/N \rightarrow \kappa \in [0, \infty)$, and for some $\delta > 0$ the centered RCT and OBS evaluation-score arrays have uniformly bounded conditional $(2 + \delta)$ moments;
- (R3) $n \text{Var}\{\hat{\theta}_r(\hat{\lambda}, \hat{\omega})\} \rightarrow \sigma^2 \in (0, \infty)$ in probability, and the combined sample-variance estimator is ratio-consistent for this variance.

Center the two evaluation-score arrays at their population means. Their weighted sum is $\hat{\theta}_r - \theta$. Condition (R2) supplies the two-sample triangular-array central limit theorem: the RCT Lyapunov remainder is of order $n^{-\delta/2}$, and the OBS remainder also vanishes when $n/N \rightarrow \kappa < \infty$. Condition (R3) and Slutsky's theorem yield the studentized limit and asymptotic coverage of the normal-approximation interval.

C.15. Proof of Theorem 3

Under either exact $r = r_0$ transport or its common-marginal special case, Equation (6) gives $\mathbb{E}_R(\Delta \mid X) = 0$. Hence $\text{Cov}_R(\Delta, \hat{g}) = 0$, as shown in Equation (89). Multiplying Equation (18) by n therefore gives

$$\tilde{V}(\lambda, \omega) = \sigma_0^2 + 2C\lambda - 2D\omega + A\lambda^2 + B\omega^2.$$

At $(\lambda, \omega) = (0, 0)$, this equals σ_0^2 . Holding $\lambda = 0$ and completing the square in ω gives the unique PPI⁺⁺-only oracle D/B and variance $\sigma_0^2 - D^2/B$. Completing the two separate squares gives the joint oracle $(-C/A, D/B)$ and variance $\sigma_0^2 - C^2/A - D^2/B$. The two variance gaps are D^2/B and C^2/A . Because $A, B > 0$, they are nonnegative and vanish exactly when $D = 0$ and $C = 0$, respectively.

C.16. Density-ratio characterizations and estimated-ratio inheritance

Proof of Proposition 4. The defining property of the Radon–Nikodym derivative gives

$$\mathbb{E}_O\{r_0(X)h(X)\} = \mathbb{E}_R\{h(X)\}$$

for every integrable h . If a nonnegative normalized function $\tilde{r}$ gives the same equality for every bounded measurable h , then $\int_A \tilde{r} dP_O = P_R(A) = \int_A r_0 dP_O$ for every measurable set A . Uniqueness of the Radon–Nikodym derivative gives $\tilde{r} = r_0$ P_O^X -almost surely.

Proof of Proposition 5. If $r_0 \in L_2(P_O^X)$, then for every $a \in L_2(P_O^X)$,

$$\begin{aligned} \mathcal{R}(a) &= \mathbb{E}_O\{a(X)^2\} - 2\mathbb{E}_R\{a(X)\} \\ &= \mathbb{E}_O\{(a(X) - r_0(X))^2\} - \mathbb{E}_O\{r_0(X)^2\}. \end{aligned}$$

The squared term is uniquely minimized at $a = r_0$ in $L_2(P_O^X)$. Restricting the balance equations to the coordinates of ϕ tests equality only against functions in their linear span, which proves the finite-feature qualification.

Proof of Lemma 1. For $a_\gamma = \phi^\top \gamma$, the gradient of the objective in Equation (32) is twice

$$\left[\mathbb{P}_O^{\text{nuis}}\{\phi(X)\phi(X)^\top\} + \rho I_p \right] \gamma - \mathbb{P}_R^{\text{nuis}}\{\phi(X)\}.$$

The stated invertibility condition makes its unique zero Equation (33).

Classifier odds. Let p_R and p_O be densities with respect to a common dominating measure. In the pooled nuisance population,

$$s_0(x) = \frac{\pi p_R(x)}{\pi p_R(x) + (1 - \pi)p_O(x)}.$$

Solving for $p_R(x)/p_O(x)$ gives

$$r_0(x) = \frac{1 - \pi}{\pi} \frac{s_0(x)}{1 - s_0(x)},$$

which proves Equation (34) in Definition 9. The same algebra with a fitted domain probability and the effective training prior yields $\hat{r}_{\text{CLS}}$.

Proof of Proposition 6. The log of a positive weighted exponential sum is convex in ξ . At an interior minimizer, differentiating the objective in Definition 10 gives

$$\begin{aligned} 0 &= \frac{\mathbb{P}_O^{\text{nuis}}\{\widehat{r}_{\text{base}}(X)e^{\widehat{\xi}^\top \phi(X)}\phi(X)\}}{\mathbb{P}_O^{\text{nuis}}\{\widehat{r}_{\text{base}}(X)e^{\widehat{\xi}^\top \phi(X)}\}} - \mathbb{P}_R^{\text{nuis}}\{\phi(X)\} \\ &= \mathbb{P}_O^{\text{nuis}}\{\widehat{r}_{\text{BAL}}(X)\phi(X)\} - \mathbb{P}_R^{\text{nuis}}\{\phi(X)\}. \end{aligned}$$

The denominator normalizes the OBS weights to one, and positivity follows from the positive base ratio and exponential tilt. A finite interior optimizer can exist only when the target feature mean lies in the relative interior of the convex hull of the weighted OBS feature vectors. Boundary or infeasible cases require a separate approximate-balance rule and are not covered by the exact first-order condition.

Proof of Proposition 7. Item 1. On the tuning OBS sample, $\widehat{r}\widehat{g} = r_0\widehat{g} + (\widehat{r} - r_0)\widehat{g}$, and the sample-variance identity $\widehat{\text{Var}}(U + V) = \widehat{\text{Var}}(U) + 2\widehat{\text{Cov}}(U, V) + \widehat{\text{Var}}(V)$ with $U = r_0\widehat{g}$ and $V = (\widehat{r} - r_0)\widehat{g}$ proves Equation (39). The moments $\widehat{A}$, $\widehat{C}$, and $\widehat{D}$ in Equation (26) do not involve the ratio, so the ratio enters Equation (28) only through $\widehat{B}$.

Item 2. At the same selected coefficients, the two estimators in Equation (13) share the term $\mathbb{P}_{R,n}Z_{\widehat{\lambda}} - \widehat{\omega}\mathbb{P}_{R,n}\widehat{g}$ and differ only in $\widehat{\omega}\mathbb{P}_{O,N}(r\widehat{g})$, whose difference across $r = \widehat{r}$ and $r = r_0$ is $\widehat{\omega}\mathbb{P}_{O,N}\{(\widehat{r} - r_0)\widehat{g}\}$.

Item 3. Condition on all fitted and selected objects. With the evaluation scores of Algorithm 1, $\widehat{\theta}_{\widehat{r}} = \mathbb{P}_{R,n}\psi_R + \mathbb{P}_{O,N}\psi_O$. The exact drift identity in Equation (15) gives the conditional mean $\theta + \widehat{\omega}\mathbb{E}_O\{(\widehat{r} - r_0)\widehat{g}\}$, and the transport identity in Equation (30) rewrites $\mathbb{E}_O(r_0\widehat{g})$ as $\mathbb{E}_R(\widehat{g})$. Independence of the two evaluation samples gives $\text{Var}(\widehat{\theta}_{\widehat{r}}) = \text{Var}_R(\psi_R)/n + \text{Var}_O(\psi_O)/N$, and each sample variance in $\widehat{V}$ is conditionally unbiased for the corresponding variance, which proves Equation (41).

Proof of Theorem 5. Write $\beta \triangleq \widehat{\omega}\{\mathbb{E}_O(\widehat{r}\widehat{g}) - \mathbb{E}_R(\widehat{g})\}$ for the drift in item 3 of Proposition 7. Continue conditioning on the fitted and selected objects. Center the two evaluation scores at their respective conditional means. The resulting two independent triangular arrays sum to $\widehat{\theta}_{\widehat{r}} - (\theta + \beta)$ and have total variance V_n . On the event $nV_n \geq \sigma^2/2$, whose probability tends to one, the uniform conditional $(2 + \delta)$ -moment bound and bounded n/N make the conditional Lyapunov ratio bounded by an $O_p(1)$ factor times

$$n^{-\delta/2} + \frac{n^{1+\delta/2}}{N^{1+\delta}} = o_p(1).$$

Thus its conditional law given the fitted and selected objects converges in probability to $N(0, 1)$ after division by $\sqrt{V_n}$. This also implies the unconditional convergence.

It remains to establish rather than assume variance consistency. Let $\eta \triangleq \min(\delta/2, 1)$. For a centered score ξ , the conditional $(2 + \delta)$ -moment bound gives $\mathbb{E}(|\xi^2 - \mathbb{E}\xi^2|^{1+\eta} \mid \text{fitted objects}) = O_p(1)$. The von Bahr–Esseen inequality therefore bounds the conditional $(1 + \eta)$ -moment of the centered empirical second moment by $O_p(m^{-\eta})$ for a sample of size m . Moreover, the squared empirical mean is $o_p(1)$ because its conditional expectation is $O_p(m^{-1})$. Consequently, for $S \in \{R, O\}$,

$$\widehat{\text{Var}}_{S,\text{eval}}(\psi_S) - \text{Var}_S(\psi_S) = o_p(1).$$

Consequently, because n/N is bounded,

$$n\{\widehat{V} - V_n\} = o_p(1) + \frac{n}{N}o_p(1) = o_p(1).$$

Together with $nV_n \rightarrow_p \sigma^2 > 0$, this proves $\widehat{V}/V_n \rightarrow_p 1$. Finally,

$$\frac{\widehat{\theta}_{\widehat{r}} - \theta}{\sqrt{\widehat{V}}} = \frac{\widehat{\theta}_{\widehat{r}} - (\theta + \beta)}{\sqrt{V_n}} \sqrt{\frac{V_n}{\widehat{V}}} + \frac{\beta}{\sqrt{V_n}} \sqrt{\frac{V_n}{\widehat{V}}}.$$

The bias condition and Slutsky's theorem prove the stated limit and coverage.

Proof of Corollary 5.1. Fix $k \in \{\text{RCT}, \text{PPI}^{++}, \text{joint}\}$ and write $(\widehat{\lambda}, \widehat{\omega}) \triangleq (\widehat{\lambda}_k, \widehat{\omega}_k)$ and $(\lambda^*, \omega^*) \triangleq (\lambda_k^*, \omega_k^*)$. Decompose

$$\widehat{\theta}_{\widehat{r},k} - \widehat{\theta}_{r_0}(\lambda^*, \omega^*) = \left\{ \widehat{\theta}_{\widehat{r}}(\widehat{\lambda}, \widehat{\omega}) - \widehat{\theta}_{r_0}(\widehat{\lambda}, \widehat{\omega}) \right\} + \left\{ \widehat{\theta}_{r_0}(\widehat{\lambda}, \widehat{\omega}) - \widehat{\theta}_{r_0}(\lambda^*, \omega^*) \right\}.$$

By item 2 of Proposition 7, the first difference equals $\widehat{\omega} \mathbb{P}_{O,N}\{(\widehat{r} - r_0)\widehat{g}\}$. Its conditional mean is the drift of item 3, which the bias condition in Theorem 5 makes $o_p(n^{-1/2})$ because $nV_n = O_p(1)$. Its conditional variance is $\widehat{\omega}^2 \text{Var}_O\{(\widehat{r} - r_0)\widehat{g}\}/N$, so Equation (46) and conditional Chebyshev make its centered part $o_p(n^{-1/2})$. For the second difference,

$$\widehat{\theta}_{r_0}(\widehat{\lambda}, \widehat{\omega}) - \widehat{\theta}_{r_0}(\lambda^*, \omega^*) = (\widehat{\lambda} - \lambda^*) \mathbb{P}_{R,n} \Delta + (\widehat{\omega} - \omega^*) \{ \mathbb{P}_{O,N}(r_0 \widehat{g}) - \mathbb{P}_{R,n} \widehat{g} \}.$$

RCT validity gives $P_R \Delta = 0$, and exact transport gives $P_O(r_0 \widehat{g}) = P_R(\widehat{g})$. Together with the second-moment bounds in Equation (45) and bounded n/N , these identities make both empirical factors $O_p(n^{-1/2})$, and coefficient consistency makes the second difference $o_p(n^{-1/2})$. Combining the two differences proves Equation (47). Convergence of the exact-ratio second moments then transfers the ordering in Theorem 3, which is Equation (48).

C.17. Counterexamples to adaptive reuse

The fixed-coefficient theorem cannot be extended by selecting a coefficient on the same evaluation outcomes without a separate analysis.

Reuse through λ . Let X be degenerate, $\theta = 0$, $Z_0 = 0$, and let the Δ_i be i.i.d. Rademacher variables. Every fixed λ satisfies $\mathbb{E}\{\lambda \mathbb{P}_{R,n} \Delta\} = 0$. If the same outcomes select $\widehat{\lambda} = \text{sign}(\mathbb{P}_{R,n} \Delta)$, then

$$\mathbb{P}_{R,n}(Z_0 + \widehat{\lambda} \Delta) = |\mathbb{P}_{R,n} \Delta|,$$

whose expectation is strictly positive.

Reuse through ω . Under the common covariate marginal, let $\widehat{g}(X)$ be a nondegenerate mean-zero Rademacher variable and let the RCT-valid score be identically zero. Every fixed ω centers $\omega\{\mathbb{P}_{O,N} \widehat{g} - \mathbb{P}_{R,n} \widehat{g}\}$. Choosing $\widehat{\omega}$ as the sign of that same bracket produces its absolute value, which

again has strictly positive expectation. The bias in both examples comes solely from adaptive reuse.